

Alzheimer's disease staging using enhanced inception-ResNet-V2 and improved XceptionNet models for 3D MRI classification and segmentation

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ABSTRACT

Background: Neurologists have a significant challenge due to the progressive nature of Alzheimer's disease (AD) and its severe effects on cognitive function. Recent advances in neuroimage analysis have opened the door to novel machine-learning techniques that could greatly improve AD diagnosis, progression prediction, and detection.

New method: In this research, we provide an enhanced hybrid deep learning approach for combined AD classification and segmentation. An enhanced inception-ResNet-V2 model is used for the multi-class classification of AD and an improved XceptionNet model is used to segment affected brain region of AD. The spatial features present in 3D MRI scans are effectively extracted by the parallel convolutional neural network (PCNN) model. **Results:** The OASIS and ADNI datasets were used in this research to detect and classify the AD stage. The proposed approach yielded consistently excellent testing accuracy and outstanding training accuracy. For testing, a higher accuracy of 99.5% for the OASIS dataset and 99.7% for the ADNI dataset is attained using the proposed approach.

Comparison with existing methods: Based on 3D MRI brain scans, these results demonstrate the exceptional ability of the proposed models, especially the Improved XceptionNet, to identify AD reliably. Based on the findings of the experiment, the proposed model outperforms cutting-edge deep learning models for classification and segmentation.

Conclusions: The experimental results show that incorporating advanced architectures significantly improves the precision of detecting and assessing brain changes associated with AD, offering practical tools for early diagnosis and monitoring the disease naturally.

1. Introduction

A collection of neurological conditions collectively referred to as dementia is marked by a deterioration in cognitive abilities like memory, thinking, and decision-making, which have a substantial impact on day-to-day activities (Hasan et al., 2024). Despite being increasingly prevalent in older persons, it is not a typical aspect of aging (Zhang et al., 2024). The World Health Organization (WHO) states that over 10 million new instances of dementia are recorded annually, with over 55

million individuals living with the disease globally (Shastri 2024; Yang et al., 2024). The most common kind of dementia is AD, accounting for 60–80% of all cases (Shankar et al., 2025). AD first manifests as moderate memory loss, but as the disease progresses, it causes significant cognitive impairment, communication difficulties, and behavioral abnormalities (Waldo-Benítez et al., 2024; Sasithradevi et al., 2024). Learning and memory-related brain regions like the entorhinal cortex and hippocampus are first affected by the illness, which then progressively spreads to other parts of the brain, causing increasingly serious

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neurological damage (Thamizharasi, Lakshmi 2025; Pallawi, Singh 2026).

Normal Control (NC), Late Mild Cognitive Impairment (LMCI), Mild Cognitive Impairment (MCI), Early Mild Cognitive Impairment (EMCI), and AD are the five phases of the development, according to a more thorough classification (Almansour et al., 2025; Sam et al., 2026). Due to small anatomical differences in brain imaging data, it is clinically difficult to distinguish between these intermediate stages (Jraba et al., 2024). Conventional neuropsychological evaluations and cognitive tests are frequently laborious and, depending on professional experience, may yield subjective results (Hu et al., 2026; Ali et al., 2026). In order to facilitate early and precise diagnosis, automated approaches utilizing machine learning and deep learning techniques have been investigated (Rao et al., 2024). Since patients can still function independently and benefit from prompt intervention strategies, early discovery at the MCI stage is especially crucial (Basheera, Satya Sai Ram 2024).

Developing machine learning and deep learning models for AD diagnosis using neuroimaging data, such as MRI, has been a major focus in recent years (Chaudhary et al., 2024). In classification tasks, convolutional neural networks (CNNs), hybrid models, and transfer learning techniques have shown encouraging outcomes. For AD detection, a number of researchers have used architectures including CNN-SVM, DEMNET, and traditional CNN models (Sasikala 2024; Hatami et al., 2024). Similarly, because of their excellent feature extraction capabilities, pre-trained models like ResNet50, VGG16, AlexNet, and InceptionNet have been widely used (Gautam, Singh 2025).

Despite these developments, existing methods continue to have a number of significant drawbacks. First, the progressive nature of AD is not adequately captured by binary or three-class classification (e.g., NC vs. AD or NC vs. MCI vs. AD), which is the main emphasis of many investigations. Because of these small changes in brain anatomy, intermediary stages like EMCI and LMCI are frequently misclassified. Second, the majority of deep learning models use extremely deep architectures, which raises memory needs, training times, and computational complexity. Because of this, they are less appropriate for real-time clinical applications where scalability and efficiency are crucial.

The absence of efficient feature representation is another significant drawback. Fine-grained spatial variations in MRI images may be difficult for traditional CNN-based models to capture, particularly when class differences are small. Furthermore, a lot of approaches use redundant or less informative features during categorization because they lack specialized feature selection or feature fusion algorithms. The robustness and generalization of the model are directly impacted by this. Additionally, a number of models experience overfitting as a result of small datasets and class imbalance, especially in multi-stage AD datasets like ADNI.

Furthermore, the majority of current research treats segmentation and classification as distinct problems. However, classification models may not be interpretable, which is crucial for therapeutic decision-making, if the damaged brain regions are not accurately localized. While some segmentation models, such as U-Net and Mask R-CNN, offer outstanding localization, they also come with high computational costs and longer inference times. The performance of current deep learning models is further constrained by optimization issues such as gradient vanishing, incorrect parameter tuning, and sensitivity to hyperparameters.

Given these difficulties, a more effective and reliable framework that can concurrently handle feature representation, computational efficiency, and classification accuracy is required. This paper proposes a hybrid deep learning model for multi-stage AD segmentation and classification using MRI data. To overcome the drawbacks of current techniques, the proposed approach combines advanced feature extraction, optimized classification, and precise segmentation.

Combining various architectures and utilizing attention processes can greatly enhance performance in medical imaging applications,

according to recent advances in deep learning. Inspired by this, the proposed study seeks to provide a thorough framework that can reliably identify all five phases of AD, including intermediate stages like EMCI and LMCI. The creation of a reliable and generalizable model is further aided by the utilization of extensive datasets like ADNI and OASIS.

Overall, this research advances the creation of an effective computer-aided diagnosis system that is better suited for practical clinical applications by increasing interpretability through segmentation and improving classification accuracy.

1.1. The major contribution of the proposed research

- To aid in the early diagnosis and detection of AD, a new model called a hybrid of the Enhanced Inception-ResNet-V2 and Improved XceptionNet model is proposed.
- To enhance image quality and improve model performance, the pre-processing stage involves noise removal using the Modified Non-Local Means (MNLN) filter, contrast enhancement through the Improved Adaptive Contrast Enhancement (IACE) algorithm, and data augmentation using conventional techniques to increase the diversity and robustness of the training dataset.
- To retrieve discriminative spatial features of critically modified microstructures in the brain, the parallel convolutional neural network (PCNN) model is developed.
- Finally, the acquired features are used to classify the multi-class AD stages using the Enhanced Inception-ResNetV2 model.
- To accurately segment the affected brain regions in AD, the Improved XceptionNet model is employed, leveraging its enhanced architecture for the precise delineation of diseased tissues.
- By contrasting the proposed deep learning models' performance with the state-of-the-art techniques based on predetermined evaluation measures, such as F1-score, recall, precision, accuracy, IOU, DSC, confusion matrix, and ROC curve, the efficacy of the models is validated.

1.2. Novelty of the proposed approach

While a number of deep learning models have been developed for the diagnosis of AD, the majority of existing research focuses on either segmentation or classification separately and frequently uses conventional architectures without tailoring them to the distinctive properties of brain MRI data. The proposed research presents a hybrid framework that integrates segmentation, classification, and improved feature extraction models into a single pipeline. This work is novel because it combines an enhanced Inception-ResNetV2 model with multi-scale depth-wise separable convolutions for effective multi-class classification, an improved XceptionNet with Swish activation and weighted loss for accurate lesion localization, and a parallel convolutional neural network for improved spatial feature extraction. This combination enables the system to detect small regional irregularities as well as global structural patterns, which are frequently missed by traditional methods. The proposed architecture improves diagnostic accuracy while preserving computational economy by simultaneously optimizing these elements, which makes it more appropriate for real-world clinical applications.

1.3. Organization of the research

The following are the contents of the paper: For AD diagnosis, 2 discusses the studies that have been conducted on the application of deep learning methods. A thorough explanation of the proposed approach used in this research is given in 3. The experiments and outcomes of the proposed models are detailed in 4, along with a discussion of the benefits and drawbacks of the proposed approach. 5 concludes this report by outlining the scope of the next investigation.

2. Literature review

As AD diagnostic tools, several recent studies on creating deep learning models have been proposed. We have examined a few recent deep learning-based studies for multi-class AD diagnosis. For the AD diagnosis, (Ravi et al., 2024) employed various CNN models. The ResNet-50V2 provides the best results for AD classification in this paper. The model with the highest accuracy in identifying the image's region of focus is used to estimate its class using visualization techniques like Saliency Map and Grad-CAM.

A CNN-based model for multi-stage AD classification was created by (Kaya, Çetin-Kaya 2024). In this paper, we present a particle swarm optimization-based technique to optimize CNN architectures for classifying the severity of AD by fine-tuning hyperparameters. A different method for AD classification was developed by (Ghassan Al Rahbani et al., 2024), which combined two CNN models, EfficientNet and ResNet, with a post-processing technique. Empirical evaluations of publicly available datasets using a weighted averaging ensemble learning approach and the complementary strengths of the CNN models, ADNI, and OASIS assess the methodology. The distinctive feature of the suggested method is the combination of many CNN designs with a specific post-processing algorithm.

For AD brain datasets, (Khatri and Kwon, 2024) created a novel technique that incorporates convolution-attention mechanisms into transformer-based classifiers, improving performance without requiring an exorbitant amount of processing power. Outstanding results are obtained by substituting lightweight multi-head self-attention (LMHSA) for multi-head attention, using inverted residual (IRU) blocks, and adding local feed-forward networks (LFFN). A deep learning-based method for multi-stage AD classification was developed by (Kumar and Maguluri, 2024). From each MRI scan image, the features are extracted using the GoogLeNet model. After that, an FFNN was used to predict the AD stage using these attributes. To guarantee reliable performance, the FFNN model, which makes use of GoogLeNet features, was rigorously trained across several epochs with small batch sizes.

A 3D-CNN-based method for AD prediction was presented by (Rahman et al., 2025), in which volumetric MRI data is processed using 3D convolutions to maintain spatial continuity across slices. Their approach showed the importance of recording inter-slice data, but the model needs more processing power and intricate training techniques. A CNN-based system utilizing transfer learning with MobileNet for multi-stage AD classification was presented by (Mohi ud din dar et al., 2023). The model relied on traditional architectures without addressing fine-grained feature differences across intermediate stages like EMCI and LMCI, although it achieved reasonable performance with lower computing costs.

In addition to AD, deep learning models have also been successfully applied to related medical imaging tasks. An automated CNN-based system for tumor classification was created in a related study by Rasheed et al. (2023), which achieved great generalization performance and high accuracy. These studies demonstrate the efficiency of optimized CNN architectures and attention processes in medical picture processing.

These techniques effectively categorize the multi-class AD stages, according to the literature review above, but they provide several training challenges. (i) Only a few hyperparameters, including optimizer, learning rate, batch size, and dropout, are often optimized in custom and transfer learning models; this is insufficient to get a successful outcome across several datasets. (ii) When there is a small number of labelled datasets, the majority of existing models suffer from over-fitting; the over-fitting issue can be resolved by maximizing the number of filters and the number of convolution layers; it is necessary to investigate the effects of architecturally optimizing CNN (iii) Existing models like, EfficientNetV2, LFFN, and ResNet have a lot of parameters because they are utilized in transfer learning. The models' training phases are time-consuming and prone to over-fitting. We create a better

hybrid deep learning-based multi-class AD classification system to address these problems.

2.1. Problem statement

There is presently no known cure for AD, which affects millions of individuals globally. Early detection of AD is still essential for efficiently managing its burden, unlike cancer, which has made progress in halting its progression. The cancer cells are effectively classified by the Mask R-CNN model using high metric values. Due to its shorter inference time, it does not provide parameter tuning. For AD classification, the Faster RCNN increases the AUC measure. However, some untrained and unlabeled data lead to a decline in performance. For the cells, the classification efficiency is optimized by the basic UNet model by focusing on pixel segmentation. Nevertheless, it has trouble with time and complexity computation. When accuracy, sensitivity, and specificity are attained, several deep learning models like ResNet, AlexNet, and VGG models improve system robustness. However, it does not allow real-time updated images and does not apply to real-world applications. The discriminative features are effectively extracted using the ANFIS classifier, and it reaches the required accuracy value. Nevertheless, when handling a high number of images, it becomes unstable. More accurate values are obtained by the Multilayer Perceptron (MLP) concerning less important features. However, due to the overlapping images, performance suffers. The majority of previous studies concentrated on either multi-class categorization for three stages (NC, MCI, and AD) or binary classification (NC vs. AD). To determine the disease's severity or MCI stage, all intermediate stages must be examined and categorized. This work addresses the limitations of existing deep learning and machine learning models for AD classification using MRI data by presenting an improved hybrid deep learning-based technique for early AD detection in a time-consuming manner.

3. Proposed methodology

The proposed research aims to use brain 3D-MRI images to detect AD. The two publicly available datasets, including ADNI and OASIS datasets, are used for collecting brain MRI images. Our main goal is to develop a method for early AD diagnosis. Several essential steps are included in the proposed approach for the hybrid deep learning model to improve the categorization of brain MRI images for early AD diagnosis and extract pertinent characteristics from MRI images. The entire research workflow, including key elements like data collection, image pre-processing, feature extraction, classification, and segmentation, is depicted in Fig. 1. Data collection with MRI equipment is the first step, and pre-processing comes next. The MNLM filter and the IACE algorithm are two examples of filtering and contrast enhancement techniques used in pre-processing.

Compared to traditional deep learning pipelines utilized in AD analysis, the proposed methodology includes a number of architectural and operational changes. This work uses a hybrid design, where each pipeline stage is optimized for a particular task, in contrast to conventional methods that use a single pre-trained model. First, a parallel CNN structure is used to extract both contextual and fine-grained information from MRI images by capturing various spatial features. An enhanced Inception-ResNetV2 model that uses multi-scale depth-wise separable convolutions to reduce parameter complexity while maintaining discriminative capabilities, then processes these features. An enhanced XceptionNet model is used for region-level analysis; Swish activation and weighted cross-entropy loss are added to increase segmentation accuracy when there is class imbalance. One of the main innovations of the proposed framework is the coordinated application of these modified architectures, which sets it apart from existing single-model or unmodified transfer learning techniques.

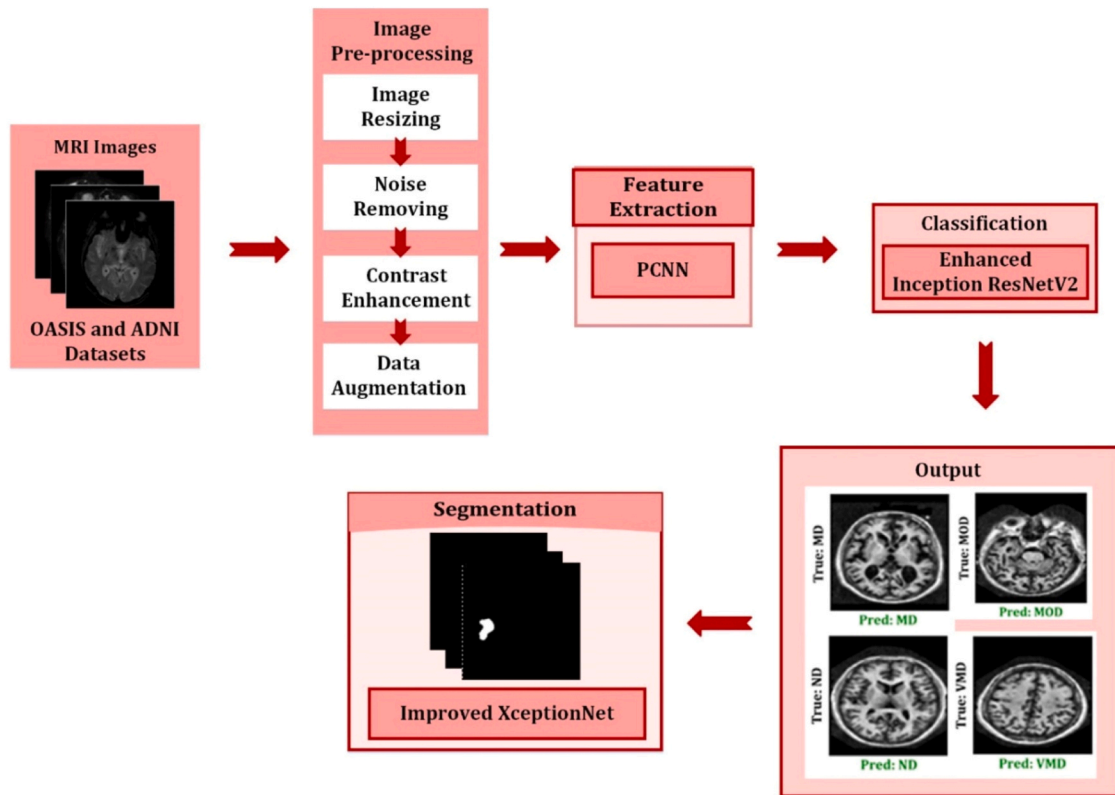


Fig. 1. The architecture of the hybrid deep learning-based proposed AD diagnosis system.

3.1. Database description

To evaluate the efficacy of our suggested approach, we used two public datasets. We obtained the data in this article from the ADNI and the OASIS database.

3.1.1. OASIS dataset

Another publicly accessible resource with neuroimaging information for AD diagnosis is called OASIS. It typically displays cross-sectional and longitudinal MRI data from patients with dementia and those who are cognitively normal. While the cross-sectional collection comprises the sagittal images of 416 patients between the ages of 18 and 96, the longitudinal category comprises numerous scans at various visits for each subject. In general, the Clinical Dementia Rating (CDR) can be used to evaluate AD risk factors. For this dataset in our investigation, a CDR score of 0 indicates that a patient has dementia, whereas a value of 0 indicates that a patient is cognitively normal. It contains 9488 images in total, representing various AD stages. This dataset has 4 classes: non-demented (ND), mild demented (MD), very mild demented (VMD), and moderate demented (MOD). Initially, the dataset shows a class imbalance problem, where the MOD class contains only 488 images, while the other three classes (ND, MD, and VMD) contain approximately 3000 images each. To perform the experiments, 80% of the dataset was used for training, and 20% was used for testing. The distribution of samples is presented in Table 1.

Table 1

The dataset distribution of the OASIS dataset.

Class Name	Training	Testing
Mild Dementia (MD)	2400	604
Moderate Dementia (MOD)	390	102
Non-Demented (ND)	2400	602
Very mild Dementia (VMD)	2400	590

To address the class imbalance issue, data augmentation techniques were applied to the MOD class in the training set. For each original image in this class, multiple augmented images were generated using conventional techniques such as rotation, flipping, and scaling. The dataset images are taken from the Kaggle website (<https://www.kaggle.com/datasets/pulavendranelvaraj/oasis-dataset>). Fig. 2 displays a selection of the OASIS dataset's images.

3.1.2. ADNI dataset

One of the main sources for model construction and assessment in this study is the AD Neuroimaging Initiative (ADNI) dataset. The dataset images are taken from the Kaggle website (<https://www.kaggle.com/datasets/yash030604/adni-png-20000-all>). The pre-processed brain MRI scans in PNG format were sourced from a publicly accessible library on Kaggle. The 21,760 images in the dataset are divided into five classes that correspond to various stages of Alzheimer's disease: AD, Cognitively Normal (CN), EMCI, LMCI, and MCI. The 4352 images in each class ensure a fair distribution across all categories.

After pre-processing, an 80:20 split ratio was used to randomly separate the dataset into training and testing sets for experimental analysis. As a result, as shown in Tables 2, 3, 482 images per class were used for training, and 870 images per class were set aside for testing. This balanced distribution guarantees equitable learning throughout all stages of the disease and aids in lowering model bias. From each class, the samples of data are displayed in Fig. 3.

3.2. Image pre-processing

The image pre-processing techniques used in this model are explained in this section. To make sure that images are compatible with the CNN architecture, preprocessing is necessary. First, each image in the collection is reduced to a consistent 226×226 pixel size. This standardization is crucial since larger images may mask crucial details needed for an accurate diagnosis. Following the resizing procedure, each

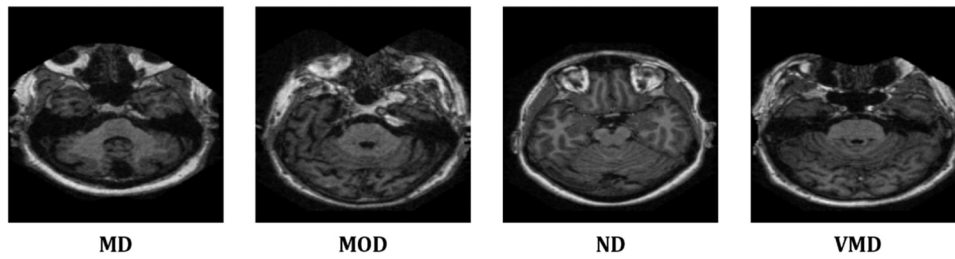


Fig. 2. The sample images of the OASIS dataset with different AD stages.

Table 2

The dataset distribution of the ADNI dataset.

Class Name	Training	Testing
AD	3482	870
CN	3482	870
EMCI	3482	870
LMCI	3482	870
MCI	3482	870

Table 3

The balanced images after applying data augmentation for the OASIS dataset.

Class Name	No of Images
MD	2400
MOD	2340
ND	2400
VMD	2400

image's pixel values are changed to fall between 0 and 1. This normalization step is crucial to enhancing CNN's ability to process the input images effectively. Following that, the images' contrast is enhanced using the IACE method, and the noise is eliminated using the MNLM filter.

The pre-processing techniques used are essential for boosting the learning capacity of the proposed deep learning models, in addition to improving visual quality. The MNLM filter for noise reduction serves to decrease unimportant changes driven by imaging errors while maintaining crucial structural information like tissue transitions and cortical boundaries. Instead of learning noise-induced features, which frequently result in unstable training and decreased generalization, this allows the feature extraction module to capture more reliable spatial patterns.

Similarly, the separability of various brain tissues is greatly enhanced by contrast enhancement utilizing the IACE method. Especially in the initial phases of Alzheimer's, the intensity differences between gray matter, white matter, and cerebrospinal fluid in MRI images are frequently seen in children. The proposed technique improves local contrast, which helps the network detect early disease changes, such as slight atrophy and textural alterations. When identifying intermediate stages like EMCI and LMCI, where visual distinctions are negligible, this is especially helpful.

Additionally, IACE's adaptive nature guarantees that enhancement is done selectively based on local image features, preventing noise over-amplification. As a result, both classification and segmentation tasks immediately benefit from better edge definition and more distinct region boundaries. Improved contrast aids in the precise localization of impacted areas for the segmentation model and improves discriminative feature learning across various phases of the disease for the classification model.

Overall, the combination of MNLM filtering and IACE not only enhances image quality but also helps the suggested hybrid framework achieve better classification and segmentation performance, more stable training, and faster convergence.

3.2.1. Noise removal

By making use of the redundancy in real images, the NLM filter eliminates noise. It looks for comparable patches in the area of a particular pixel and calculates a weighted average of the pixel intensities. For each pixel, a fixed-size search window is employed by the conventional NLM denoising algorithm. Pixels in an image, however, can be found in both smooth and non-smooth regions. Similarly, when compared to the original image, the filtered region's variance will be substantially different if a short search window is used to filter the smoothed area. As a result, we must examine each pixel's regional features, and the search region's predicted variance determines the optimal search window size. The most effective window size was adaptively selected using the proposed approach: large (l), medium (m), and small (s). Finding the noise variance requires estimating the noisy image to attain each region's image features accurately. Based on local intensity gradients, we have employed varying patch sizes. Smaller patches are used in high-gradient regions (such as those close to tissue borders) to maintain edges. The intensity values are skewed by Rician-distributed noise, which should be addressed using a bias correction step.

Compute the correct pixel value:

$$I_{Corrected} = \sqrt{\max(I_{NLM}^2 - 2\sigma^2, 0)} \quad (1)$$

Where the noise variance is described as σ^2 . After NLM filtering, the intensity is described as I_{NLM} .

Provide a similarity function that is structure-aware and incorporates gradient and intensity data:

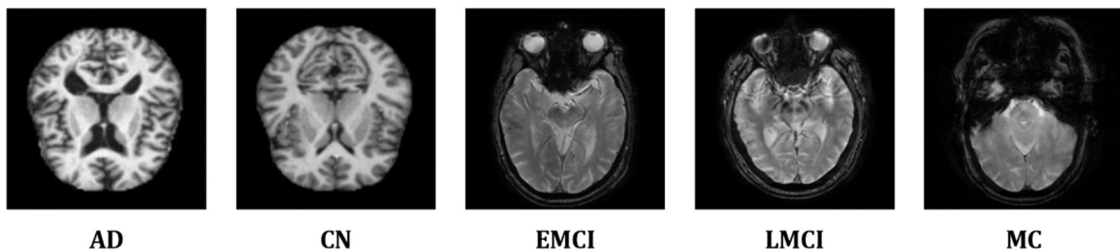


Fig. 3. The sample images of the ADNI dataset with different AD stages.

$$w(i, \leftrightarrow j) = \exp \left(-\frac{\|P_i - P_j\|_2^2}{h^2} - \alpha \frac{\|\nabla P_i - \nabla P_j\|_2^2}{\beta^2} \right) \quad (2)$$

Where the patches are described as P_i and P_j , the filtering parameter is described as h , the patch gradient is depicted as ∇P_i , and the gradient information significance is controlled by α and β . Then, to minimize computational cost and effectively search for comparable patches in larger neighborhoods, we have employed a quick search technique such as PatchMatch.

Tissue boundaries and other crucial components are preserved due to the adaptive patch size. This algorithm is the bias correction step that guarantees more accurate denoising by reducing the bias caused by Rician noise. The filtering process is greatly accelerated by the fast search techniques. The similarity metric's robustness is increased by including both intensity and gradient information, which also improves filtering quality.

3.2.2. Contrast enhancement

The contrast of the images is improved using the Improved Adaptive Contrast Enhancement (IACE) algorithm. Better visibility in regions with minutely intensity changes, like gray and white matter boundaries or small lesions, is ensured by the IACE, which adjusts the contrast enhancement based on the local intensity distribution. By adjusting the contrast exclusively in relevant regions, the algorithm prevents over-enhancement and preserves anatomical structures without adding artifacts.

The adaptive contrast enhancement method modifies contrast enhancement strength according to local features, in contrast to conventional global contrast enhancement algorithms. Within each gradient level, the adaptive contrast enhancement technique is used in conjunction with correcting the local gray level information to boost the image's dynamic range. $x(r, c)$ is the center of the image window size, which is set to $(2n+1) * (2n+1)$. The following equations represent each gradient level's local mean and local variance,

$$\mu_x(r, c) = \frac{1}{(2n+1)^2} \sum_{k=r-n}^{r+n} \sum_{l=c-n}^{c+n} x(k, l) \quad (3)$$

$$\sigma_x^2(r, c) = \frac{1}{(2n+1)^2} \sum_{k=r-n}^{r+n} \sum_{l=c-n}^{c+n} [x(k, l) - \mu_x(r, c)]^2 \quad (4)$$

Where the window region has the mean value $\mu_x(r, c)$ is centred on the pixel point $x(r, c)$. On pixel $x(r, c)$, the window region's variance $\sigma_x^2(r, c)$ is centred. The window region's total number of pixels $(2n+1)^2$ is centred at the pixel point $x(r, c)$. When $x(k, l) - \mu_x(r, c)$ representing the high-frequency region, the background portion can be used to approximate the mean value. For the high-frequency region, the image's details can be improved by multiplying the gain factor. In this research, we adjusted the adaptive contrast enhancement-based gain factor calculation approach to make The local mean-square error inversely proportional to the gain factor. The following is a representation of this processing:

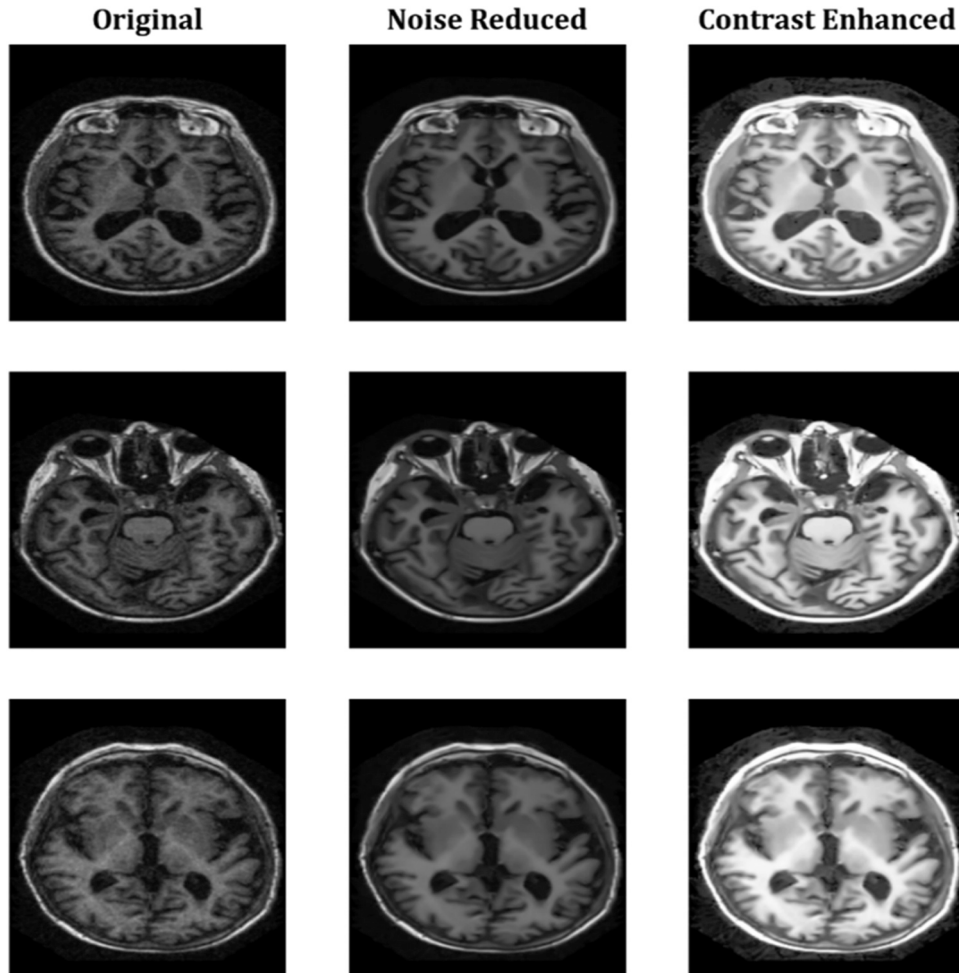


Fig. 4. The resulting images after pre-processing, the coherent image is shown in the first column, the images with noise reduced using MNLM Filtering are shown in the second column, and the contrast-enhanced image using IACE is shown in the third column.

$$f(r, c) = \mu_x(r, c) + \frac{D}{\sigma_x(r, c)} [x(r, c) - \mu_x(r, c)] \quad (5)$$

In the image's high-frequency region, the gain factor value is lower, and the local variance is greater. To prevent the gain factor from being excessively high, the global mean-square error is utilized in this research as the value of parameter D. The image's contrast is improved during IACE processing, producing a new gray-level mapping. Fig. 4 displays a sample of pre-processed images from the OASIS and ADNI datasets.

3.2.3. Data augmentation

To create the class balance dataset, the data augmentation phase is required. Additionally, this phase prevents the model from over-fitting. Usually, geometric augmentations are combined to create new, enhanced MRI brain images. There are fewer samples in the MD class in the OASIS dataset. We choose to use traditional data augmentation techniques to enlarge the images in that class to address the issue of class imbalance. The most effective method for creating a classification model that can be used to train the present data into additional data is data augmentation. The datasets can only contain a certain number of images for testing and training. Creating duplicate data and adding it to the training set is one method to address this problem. For some classification tasks, which mostly assume data augmentation, it makes sense to create duplicate data. Though the dataset is small in actuality, the target application may contain asymmetrical conditions such as position, scale, brightness, disparate orientations, etc. The CNN is trained with additional adjusted data to accommodate these scenarios. The following features have been modified for augmentation:

- The rotation range is in degrees (0–180), and the images rotate at random within that range.
- A portion or entire image may randomly translate horizontally or vertically within the ranges of height_shift and width_shift.
- Shear_range is for applying shear transformations at random.
- Zoom_range is used to apply zoomed-in images at random.
- Horizontal_flip allows a half image to be randomly flipped horizontally (without any horizontal symmetry).
- Fill_mode is used to fill in the newly generated pixels following a shift or rotation.

Both the original and enhanced images are displayed in Fig. 5.

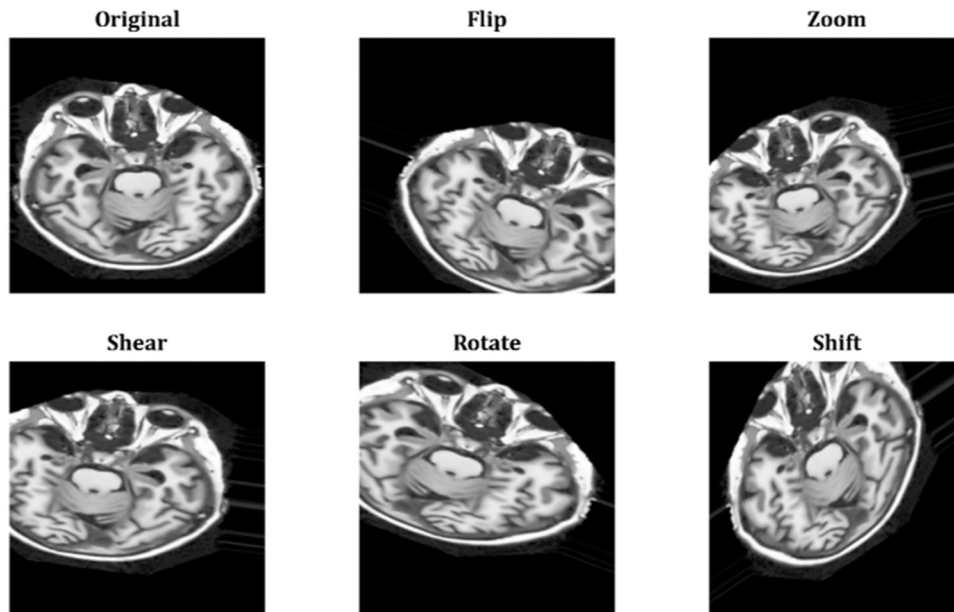


Fig. 5. Original and augmented images.

Features like rotation range (15 °), sheer transform, height shift, width shift, zoom in/out, and flip are used to enhance the original image's ratio to 1:6. As a result, the number of training samples in the MOD class was increased to a level comparable with the other classes. The balanced dataset after augmentation is shown in Table 3. This balancing step lessens the bias toward majority classes and improves the model's ability to learn characteristics from all classes. Using both original and enriched data enhances the model's capacity for generalization and promotes improved classification performance across various dementia stages.

3.3. Feature extraction

This study's primary goal was to create a CNN with fewer layers and parameters, which reduces the processing time while preserving the most noticeable features. Accurate detection of the DR levels was achieved with the Enhanced Inception ResNetV2 model due to the significant features. To get the greatest features, the convolutional layer in CNN was essentially positioned consecutively. For example, choosing too few convolutional layers could cause the model to over-fit, while choosing too many convolutional layers could cause some discriminant features to be lost. Therefore, it was necessary to select the proper number of convolutional layers to extract the most pertinent features. Six convolutional layers were chosen for this research to minimize over-fitting and retrieve the salient features. The architecture of the light-weight parallel CNN is shown in Fig. 6.

Four convolutional layers were arranged in parallel in the light-weight parallel CNN, which reduced processing time and parameters. Since the four convolutional layers were operated simultaneously, they functioned similarly to four convolutional layers but could be regarded as a single convolutional layer. Each convolutional layer has a size of 64. The activation function was ReLU, and the kernel sizes of 9×9 , 7×7 , 5×5 , and 3×3 are the first, second, third, and fourth convolutional layers, respectively. To verify the border element, the padding size in the first four convolutional layers of this research was maintained at the same level. Since the border element may occasionally contain significant information in the FIs that were examined, it is necessary to utilize the same padding. These parallel convolutional layers' output was then concatenated and sent into the sequential CNN. With a kernel size of 3×3 , the final two convolutional layers have sizes of 32 and 16, respectively. The remaining convolutional layer's padding size was maintained as valid. After each convolutional layer, a max-pooling

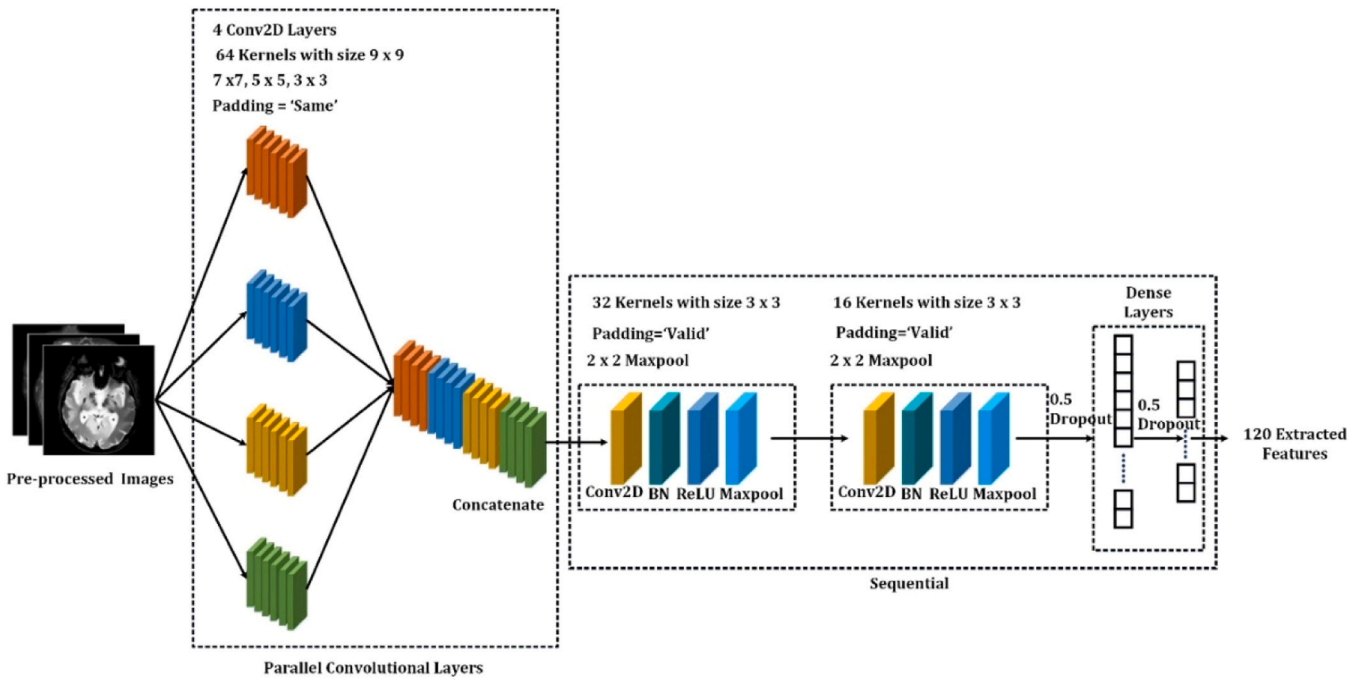


Fig. 6. The structure of the lightweight parallel CNN model for feature extraction.

layer, activation, and batch normalization are applied. By finding the greatest value in each region at the convolutional layer, max-pooling using 2×2 filters was utilized to extract the most significant regions of the FIs. Feature extraction was done from the final fully connected (FC) layer of two. With a 0.5 probability, two dropouts were used: one following the first fully connected layer and another following the last convolutional layer. Dropout, which randomly skipped 50% of all nodes, was employed to decrease over-fitting and expedite the training process. With a learning rate of 0.001 and the ADAM optimizer, the CNN model was trained for 50 epochs with a batch size of 64 to extract the features. Sparse categorical cross-entropy was used to handle the loss. A trial-and-error procedure was used to choose 120 features in total from the final fully connected layer.

3.4. Classification

To classify the multi-class AD stages, an Enhanced Inception ResNetV2 model is effectively used. While avoiding the issues of gradient explosion and disappearance, it enhances network efficiency and increases network depth. By splitting down the convolution kernel, the network's nonlinearity is strengthened, its depth is further expanded, and its processing capacity is enhanced. The Inception-ResNet-V2 model serves as the foundation for this paper. Fig. 7 displays the model's general structure as well as its module structure diagrams. Fig. 8 displays the module structure diagrams for Inception-Resnet-A, Inception-Resnet-B, and Inception-Resnet-C.

Three related modules make up the Inception-ResNet-V2 model's feature extraction module. A tensor connection is used to represent the filter concat. After the connection, the total channel's range is equal to the previous channel's range. From left to right, the convolution kernel size of the Fig.'s rectangle is first described, and then it is from top to bottom. The step length, channel number, and convolution or pooling operation are often set to 1 mark. The default filling is SAME, and V stands for valid, not filling. When compared to the Inception-ResNet module's symmetric convolution kernels, the Inception-ResNet-B and Inception-ResNet-C module's asymmetric convolution kernels are 1×7 , 7×1 , 1×3 , and 3×1 . This can deepen the network's nonlinearity and reduce calculation time during parameter operation. Following a

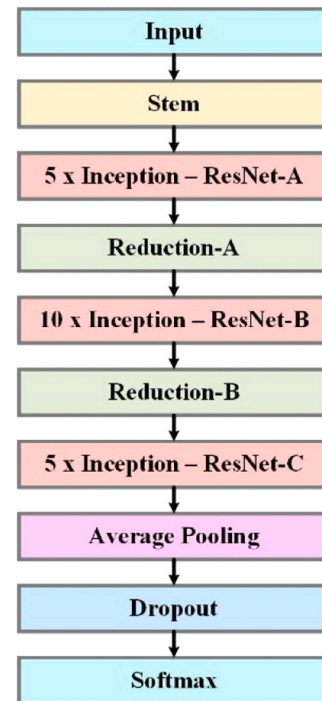


Fig. 7. Inception-ResNetV2 overall network structure.

sequence of tensor connection and convolution computations, the ReLU activation function and batch normalization processing activate the feature graph. Ultimately, the outcome of each module is obtained. Following completion of all modules, the outcome is categorized using the softmax function and a fully connected layer.

Instead of using a standard convolution kernel, a multi-scale depth-wise separable convolution kernel is used in the proposed Enhanced Inception-ResNetV2 model, significantly increasing the model's robustness. Traditional convolution techniques require a lot of computations and a lot of parameters. Instead of using a conventional convolutional

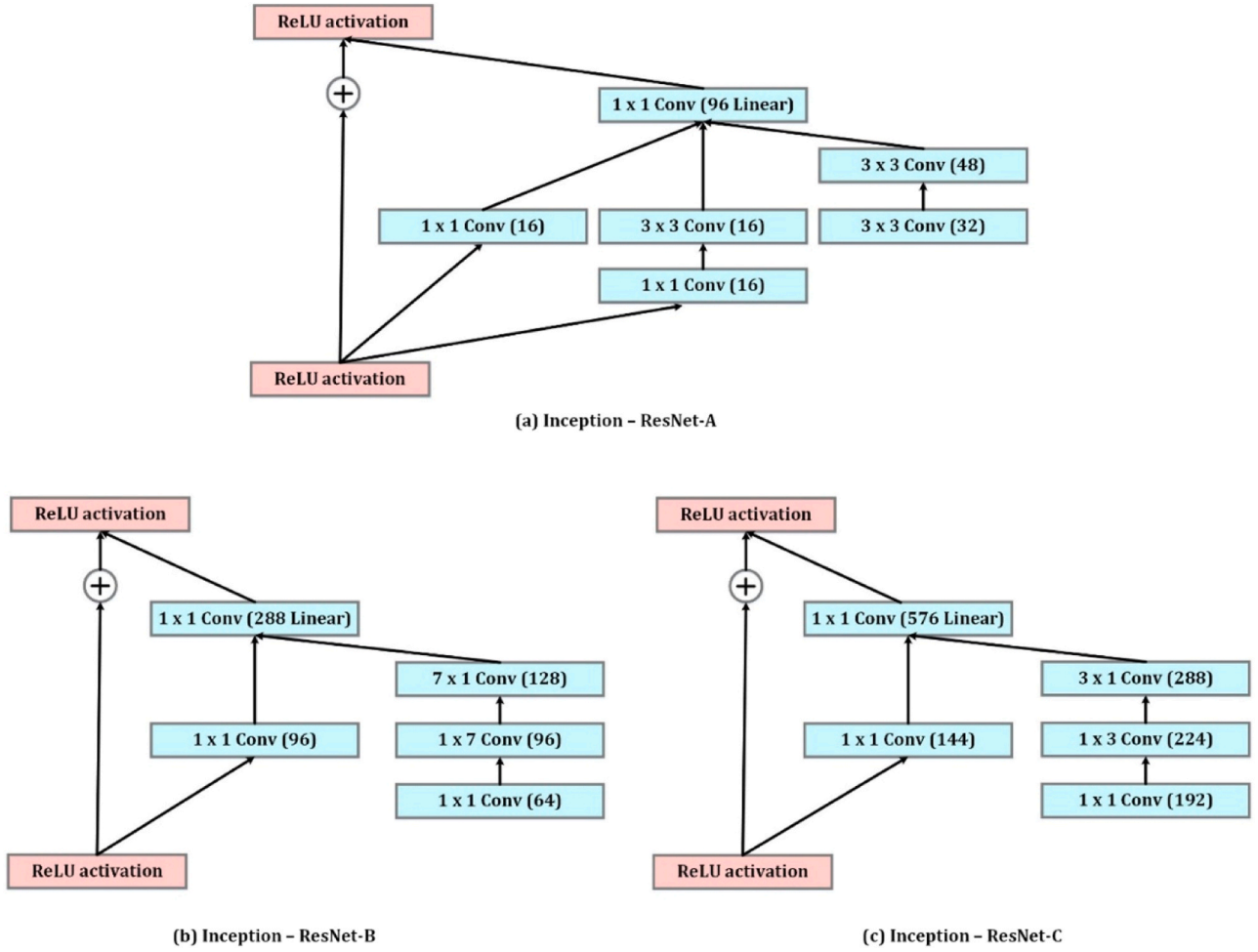


Fig. 8. Diagrams of the Inception-ResNetV2 model's module structures.

kernel to address this problem, we have employed a depth-wise separable convolution kernel, which is divided into two components: point-wise (channel mixing) and depth-wise (spatial).

3.4.1. Multi-scale depth-wise separable convolution

The operation speed and number of parameters are reduced by the depth-separable convolution. Convolution parameters can be decreased using depth-wise convolution, and the output feature graphs can be changed using point-wise convolution. In contrast to traditional convolution, which uses 3×3 channels, depth-wise convolution can reduce the convolution parameter and increase efficiency. However, it must be used in conjunction with point-wise convolution to ensure that there are exactly as many output channels as input channels. One common type of convolution procedure is point-wise convolution. The usage of the $1 \times 1 \times M$ convolution kernel is the most important component, the quantity of feature maps that must be acquired is M . To obtain the required number of output channels, the dimension is reduced by using a 1×1 convolution kernel through a point-wise convolution process. Additionally, the map from the previous phase is combined in a weighted method in the direction of depth.

The three images of one channel are formed by separating the three channels' visuals during the profoundly separable convolution process. Three feature maps are then created by independently convolving the three images, and then using the 1×1 convolution to combine them into the final eigenvalue. In this research, the deep separable convolution is the basis for the multi-scale deep separable convolution suggested. For feature graph validation, several convolution sizes are separated from the deep separable convolution in conjunction with the

Inception structure concept. In the width range, more representative features are created by selecting and combining each convolution kernel's feature graphs, all the while lowering the model's parameters and increasing efficiency. In this work, one pooling and three convolution operations make up the multi-scale deep separable convolution implemented. Deep separable convolution is the name of the three convolution techniques, and the convolution kernel has a size of 1×1 , 3×3 , 5×5 , and so on. Combining the results of the four processes with the input pooling operation's output will execute the channel's feature splices. Fig. 9 illustrates the depth-wise separable convolution procedure on multiple scales.

In conventional convolution, the more feature maps there are, the higher the convolution parameter, and the primary parameter equation is:

$$G_{(k-i)/s+1, (l-j)/s+1, n} = \sum_{i,j} K_{i,j,m} * C_{k,l,m} \quad (6)$$

The convolutional kernel is described as K , the convolution kernel size is described as i, j , the number of channels' size is described as m , the input characteristic graph is described as C , the characteristic graph size is represented as k and l , the output characteristic graph size is described as G , $(k-i)/s+1$ and $(l-j)/s+1$ is the size, the step size is described as s , and the output channels range is described as n . For conventional convolution, the number of parameters is described as $i * j * m * n$.

The primary equation of Depth-wise convolution is:

$$G_{(k-i)/s+1, m} = \sum_{i,j} K_{i,j,m} * C_{k,l,m} \quad (7)$$

The degree of correspondence between the input feature graphs' and

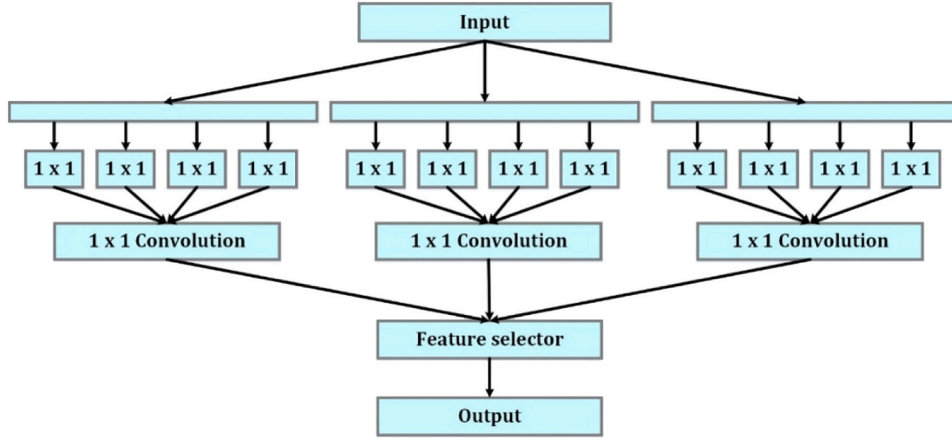


Fig. 9. The multi-scale deep separable convolution process.

output feature graphs' channel numbers is where this formula differs from conventional convolution. Since the number of parameters and the number of output channels are closely correlated, parameter computation can be simplified here.

The point-wise convolution parametric formula is:

$$G_{k,l,n} = \sum_{m,n} K_{1 \times 1 \times n} * G_{k,l,m} \quad (8)$$

With $1 \times 1 \times n$, the convolution kernel is K . The output channel's number is described as n . The $i \times j \times n$ is the parameter's range. The overall number of parameters is reduced, and efficiency in the output feature graph is increased by concatenating the Depthwise convolution with the 1×1 convolution.

3.4.2. Feature selector

The feature selector in this research is another improvement. With channel number 1, the deep separable convolution can produce multiple feature graphs, where the number of input channels is depicted as n . It is possible to achieve the outcome of deep separable convolution by combining several feature graphs. The feature selector employed in this paper, however, should display the relative importance of the several feature graphs using different weight coefficients, as each feature graph in the results has different feature values. The input and output characteristic diagram's channel number is shown by A and B , and the channel's importance is described as p . Each feature graph's global features are obtained by the feature selector first pooling each feature graph globally using depth-wise separable convolution. The significance of each feature graph can be represented by the global features. The global feature's probability is then determined using the softmax function, and each feature graph's weight probability is obtained. The final feature graph is obtained by multiplying the weight by the corresponding feature graph. With additional features, the feature map is maintained by the feature selector, which can effectively convey the image's global information. Simultaneously, it eliminates the image's unnecessary details and minimizes computation.

There is a feature selector that can choose every convolution result following multi-scale depth-wise separable convolution. The majority of the image's information is retained in the feature image that is produced. For dimensionality reduction, following the last 1×1 convolution layer, feature extraction yields positive results. The model parameters are significantly reduced while preserving the data integrity, all feature graphs are transformed into global features via the feature extractor, estimate the global feature's probability using the soft-max function, and the small global feature's feature graphs. The multi-class classification of AD diagnosis is done using the categorical cross-entropy loss function.

3.4.3. Optimization

An extension of SGD, each parameter's learning rate is modified by the Adam optimizer by utilizing the gradient's first- and second-order moment estimation. Its benefit is that, following bias correction, each iteration's learning rate has a specific range, making parameters more stable. This paper's Enhanced Inception-ResNet-V2 model makes use of the Adam optimizer. The model is optimized using the Adam optimizer to manage better the non-stationary targets and the sparse gradient of the experiment.

3.5. Segmentation

One of the well-known and powerful CNN models, the Xception architecture is developed under several key concepts, including inception module, residual connections, depth-wise separable convolution layer, and convolutional layer. Additionally, the CNN design for the activation function is crucial, and the standard activation function was developed using Swish, a new activation function. Based on the Swish image classification model, a Swish activation function is used in this research to enhance the Xception for the first melanoma diagnosis. Additionally, the weighted loss function rather than the cross-entropy loss function is used to overcome the issue of an uneven sample distribution.

According to the description, the Inception module serves as the foundation for the Xception theory. It creates spatial connections and cross-channel correlations in CNN feature maps that are completely decoupled. Through the division of input data into four categories, Cross-channel correlations are essential to the Inception model, then using a convolution of size 1×1 to average pooling and mapping correlations before sending the data to the concatenation layer. While injecting to the concatenation, output channels from overlapping are prevented by this network by using only one size of 1×1 convolution to generate convolution sizes of 3×3 without average pooling. Compared to the Inception module, which uses fully decoupled maps to operate correlations of cross-channels and spatial connections, this module is more dependable, robust, and consistent. The steps for the Xception module are described in full below:

3.5.1. Convolutional layer

The input data parts of the convolution kernels have been separated to create feature maps. Since the k^{th} layer of the feature map indicates the l^{th} , the location (i, j) upon feature value is the absolute result of the many convolution kernel that produce the feature maps, i.e.,

$$S_{i,j,k}^l = Bv_k^l + Wv_k^l C_{i,j}^l, \quad (9)$$

Where the weight vector is described as Wv_k^l . For l^{th} layer, the k^{th} filter's bias value is described as Bv_k^l , and the l^{th} layer's position (i, j) 's input

patch center is described by C_{ij}^l . The $S_{ij,k}^l$ feature map was shared in order to construct the W_k^l kernel.

The $S_{ij,k}^l$ feature map was shared in order to construct the W_k^l kernel. This procedure makes things easier and builds the network for training models in a smooth manner. The Xception module's convolutional layers are inserted using batch normalization and ReLU activation functions.

$$RELU = \begin{cases} x, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad (10)$$

The non-linear features are detected using CNN in the network non-linearity, which does not mathematically complicate the ReLU activation function, resulting in improved predictions with less over-fitting and faster convergence.

3.5.2. Depth-wise separable convolution layer

The majority of the Xception modules are contained in the depth-wise convolutions. When the model parameters are prepared in-depth, and the spatial dimensions of color channels are known, this can reduce the calculation. The depth-wise convolution creates a feature map to define $DF \times DF \times M$ by filtering the M input data set channels.

$$\hat{G}_{k,p,m} = \sum_{i,j,m} \hat{K}_{ij,m} \times F_{k+i-1,p-j-1,m} \quad (11)$$

The input feature map is F , and the output feature map alternatives are explained by $\hat{G}$ and the depth-wise convolution kernel is described as $\hat{K}$. F is a channel, which is using the filter number m in $\hat{K}$ in order to estimate the feature map's output. The image is then displayed in several color channels, each of which can be captured. The output is then fed into the next layer using the 1×1 convolution filters. Batch normalization is used after the depth-wise separable convolution layer, and the computational complexity is subsequently reduced by employing the max-pooling layer.

3.5.3. Residual connection

The ResNet architecture, in which the internal network makes identity shortcut connections straight into the last layers, has been used to achieve the residual connection. The following descriptions explain the residual block by considering the parameters p_i :

$$v_o = v_i + f(v_i, [p_i]) \quad (12)$$

Where the layer's input and output vectors are described as v_i and v_o . In residual shortcut connections, a late layer can be directed by X 's input using an identity block shortcut. Data was sent to a late layer via a 1×1 convolution operation with a 2×2 step. With 36 convolutional layers, a network is used in Xception to generate feature extraction. Except for the first and last modules with residual connections between them, it produces 14 modules. The input image in the Xception pre-trained network should be $299 \times 299 \times 3$.

3.5.4. Swish activation function

The segmentation results are efficiently produced by this activation function. Stated differently, the model's effectiveness is developed by the Swish activation function instead of the conventional ReLU activation function. The Swish activation function is expressed mathematically as follows:

$$S = m \times \text{sigmoid}(\alpha \times m) \quad (13)$$

Where an adjustable per-channel parameter is described as α , the input data is defined as m , and the sigmoid function evaluation is described as $\text{sigmoid}(\alpha \times m)$.

3.5.5. Loss function

The loss function, which is weighted cross-entropy, is used to address

the issue of unbalanced category distribution, and each category's weight value for the loss function is selected to enhance the model's capacity to learn from a limited number of examples.

The log-likelihood loss function is another name for the cross-entropy loss function.

According to the conventional cross-entropy loss function formula,

$$\text{loss}(x_{\text{class}}) = -\log \frac{e^{x_{\text{class}}}}{\sum_j e^{x_j}} = -x_{\text{class}} + \log \left(\sum_j e^{x_j} \right) \quad (14)$$

The weighted cross-entropy loss function's formula is:

$$\text{loss}(x_{\text{class}}) = -W_{\text{class}} \times \left(-x_{\text{class}} + \log \left(\sum_j e^{x_j} \right) \right) \quad (15)$$

Fig. 10 depicts the Improved Xception network's structure. The modules in this are similar to those in the original Xception; the only difference is that the Swish activation function position has been used in place of the ReLU function, it comes after global average-pooling and before logistic regression. Furthermore, we have employed the weighted cross-entropy as the loss function to address the network's over-fitting problem.

4. Result and discussion

The experimental results and thorough analysis of our tested models with statistical validation are presented in this section. Additionally, we investigated and compared the proposed models with the existing baseline models.

4.1. Experimental settings

A workstation equipped with a quad-core Ryzen 7 processor, an NVIDIA GeForce GTX 6 GB GPU, 16 GB of RAM, and Windows 10 was used to train the deep learning model. The deep learning library for Python. For tasks involving model building, training, performance assessment, and visualization, Keras with Tensorflow as the backend was utilized, along with sklearn, seaborn, matplotlib, and additional libraries.

4.2. Parameter settings

Deep learning models' performance and capacity for generalization are greatly enhanced by hyperparameter modifications. Instead of using fixed empirical selection, a systematic experimental procedure was used in this research to discover the ideal hyperparameters for the proposed models. Initially, based on frequently used settings documented in recent literature, a range of values for important parameters like dropout rate, number of epochs, batch size, and learning rate were investigated.

A trial-and-evaluation approach was used for the tuning procedure, in which several hyperparameter combinations were evaluated, and the optimal configuration was chosen based on validation performance. The models were trained iteratively while tracking validation accuracy and loss once the dataset was split into training and validation sets. Early validation trend observations were taken into account, and dropout regularization was used to prevent overfitting.

Values between 0.0001 and 0.001 were investigated for the learning rate. A learning rate of 0.001 was shown to accelerate convergence without creating training instability. Higher learning rates caused oscillations in loss, while lower learning rates caused slower convergence. Because of its robust convergence behavior and adaptable learning capabilities, the Adam optimizer was chosen, especially for medical image processing applications involving complex feature distributions. 16, 32, and 64 batch sizes were assessed. A suitable balance between convergence stability and computing efficiency was achieved with a batch size

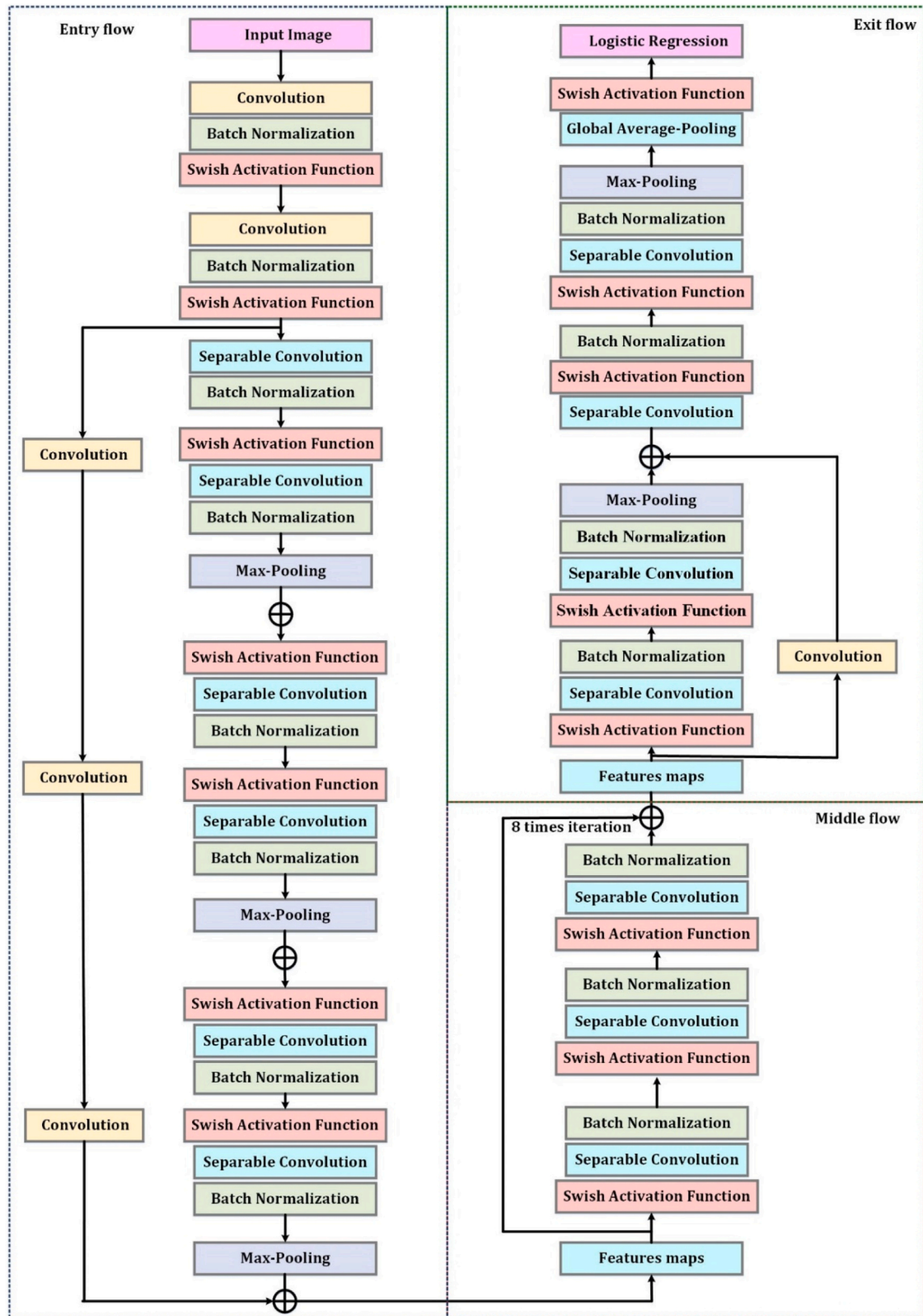


Fig. 10. The architecture of the improved Xception network.

of 32. While larger batch sizes needed more memory and did not appreciably improve performance, smaller batch sizes lengthened training times.

There was a range of 50–150 epochs. The models were found to perform steadily for around 100 epochs, after which there was no

discernible improvement, and overfitting tendencies started showing up. Consequently, the perfect settings were determined to be 100 epochs.

To lessen overfitting, the dropout rate was adjusted between 0.2 and 0.5. It was discovered that a dropout value of 0.25 was useful for

enhancing generalization while preserving the model's ability to learn. Activation functions like ReLU and Swish were also assessed; because of its smooth non-linearity, Swish performed better in the Improved XceptionNet model.

For the loss functions, the Improved XceptionNet model employed weighted cross-entropy to address class imbalance in segmentation tasks. In contrast, the Enhanced Inception-ResNetV2 model used categorical cross-entropy for multi-class classification. Table 4 summarizes the final optimized hyperparameters that were utilized to train the proposed models.

4.3. Evaluation measures

The following metrics were chosen to confirm the method's efficacy: F1-score, accuracy, precision, and recall. These metrics are frequently used to assess how well medical image classification techniques function; for medical images, sensitivity shows the likelihood of correctly detecting lesions, while specificity shows the probability of preventing misdiagnosis. It is defined as follows:

The accuracy is the ratio of the number of samples that the model properly classifies to the total number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (16)$$

The precision is the ratio of the number of samples that the model determined to be positive to the total number of positive samples.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (17)$$

The probability of truly being diseased and accurately identifying oneself as such based on the model screening criteria is known as the recall.

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (18)$$

The harmonized average of accuracy and sensitivity is known as the F1-score, and it is capable of thoroughly assessing the network model's classification effect.

$$\text{F1-Score} = \frac{2 \times \text{Sensitivity} \times \text{Precision}}{\text{Sensitivity} + \text{Precision}} \quad (19)$$

The percentage of correctly detected negative samples to all true negative samples is known as specificity.

$$\text{Specificity} = \frac{TN}{TP + FP} \quad (20)$$

The agreement between the actual and anticipated classifications is measured using Cohen's kappa score, which takes chance into account. Particularly in multi-class classification situations, it offers a more dependable assessment than accuracy.

$$k = \frac{P_o - P_e}{1 - P_e} \quad (21)$$

Where P_o stands for the observed agreement, which is equivalent to accuracy, and the expected agreement by chance is represented by P_e , which is calculated as:

$$P_e = \frac{(TP + FP)(TP + FN)(FN + TN)(FP + TN)}{(TP + FP + TN + FN)^2} \quad (22)$$

Where, the total number of accurately categorized positive samples is known as TP (True Positive), The number of positive samples that the model incorrectly classified as negative is known as FN (False Negative), The total number of correctly categorized negative samples is known as TN (True Negative), The number of negative samples that the model mistakenly considered positive is known as FP (False Positive).

Two distinct assessment metrics, Intersection Over Union (IoU) and Dice Similarity Coefficient (DSC), are used to examine the segmentation results of the proposed model. The overlap index between the ground truth and the segmented tumors obtained with the proposed framework is known as the DSC. The Jaccard Index, another name for the IoU measure, is the ratio of the area of overlap to the area of union between the ground truth and the predicted segmentation.

$$\text{DSC} = \frac{2|A \cap B|}{|A| + |B|} \quad (23)$$

$$\text{IOU} = \frac{|A \cap B|}{|A \cup B|} \quad (24)$$

4.4. Experimental results

The classification performance results of the proposed model for the OASIS and ADNI datasets are shown in Fig. 11 and Fig. 12, with particular attention paid to metrics such as F1-score, specificity, recall, precision, accuracy, and kappa score. The higher accuracy of 99.5% and 99.7% is achieved by the proposed classification model (Enhanced Inception ResNetV2) for the OASIS and ADNI datasets. It shows that the proposed model may distinguish between several AD phases with greater accuracy using performance metrics, resulting in a more accurate and dependable diagnosis. On average, the proposed model obtains 99% precision, 99.57% recall, 99.37% specificity, and 99.52% F1-score for the OASIS dataset and 99.70% precision, 99.48% recall, 99.51% specificity, and 99.68% F1-score for the ADNI dataset. Recall and precision levels are significant for being consistent, indicating strong performance in accurately recognizing true positives and true negatives. Furthermore, the low FNR and FPR show that the models can successfully reduce classification errors. This precision might make the diagnostic process more patient-friendly by lowering the requirement for intrusive diagnostic procedures or extra testing.

Cohen's kappa coefficient was used to assess the degree of agreement between the true and predicted class labels. With a kappa value of 99.30% on the OASIS dataset, the proposed model demonstrated nearly perfect agreement in each class.

Similarly, the model achieved a kappa value of 99.60% for the ADNI dataset, indicating very consistent classification accuracy even in multi-stage disease categorization.

These strong kappa values attest to the proposed model's robustness across various AD phases and its ability to produce accurate predictions that go above chance-level agreement.

4.5. Segmentation results

The brain tumor region is correctly predicted by the proposed Improved XceptionNet model using MRI images. When producing predictions using a trained model, assigning the proper label to every pixel in the segmentation map and precisely segmenting each of these areas in the image are the major goals of the proposed segmentation model. The model's predictability is evaluated using two distinct metrics, such as DSC and IOU. For example, Fig. 13 displays the results of the

Table 4

The proposed network models' optimized hyperparameters.

Hyper-parameters	Enhanced Inception ResNetV2	Improved XceptionNet
Image size	226 × 226	226 × 226
Learning rate	0.001	0.001
Optimizer	Adam	Adam
Dropout rate	0.25	0.25
Number of epochs	100	100
Batch size	32	32
Loss function	Categorical cross-entropy function	Weighted cross-entropy function

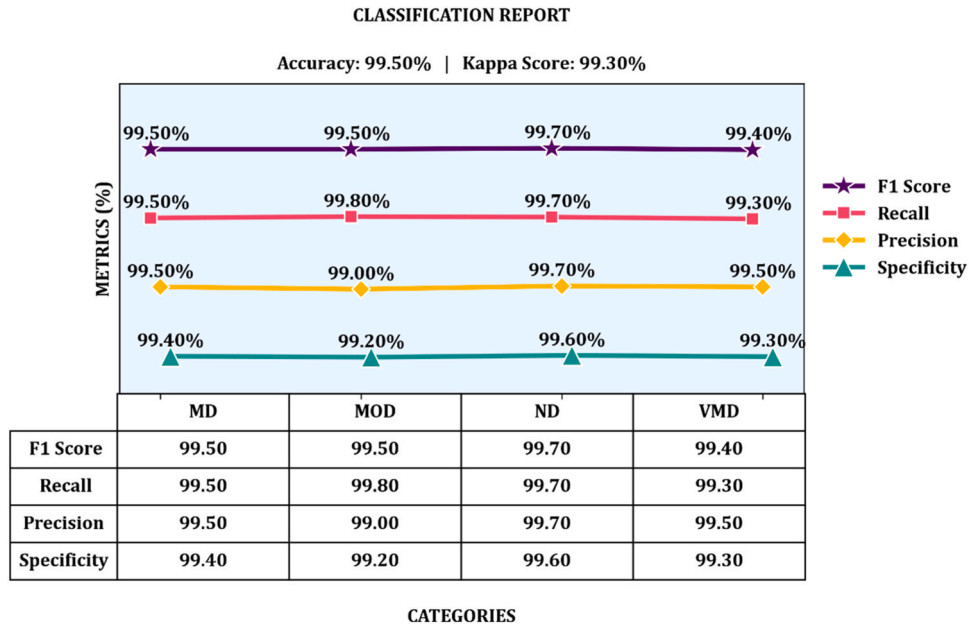


Fig. 11. The classification report of the proposed model on OASIS dataset.

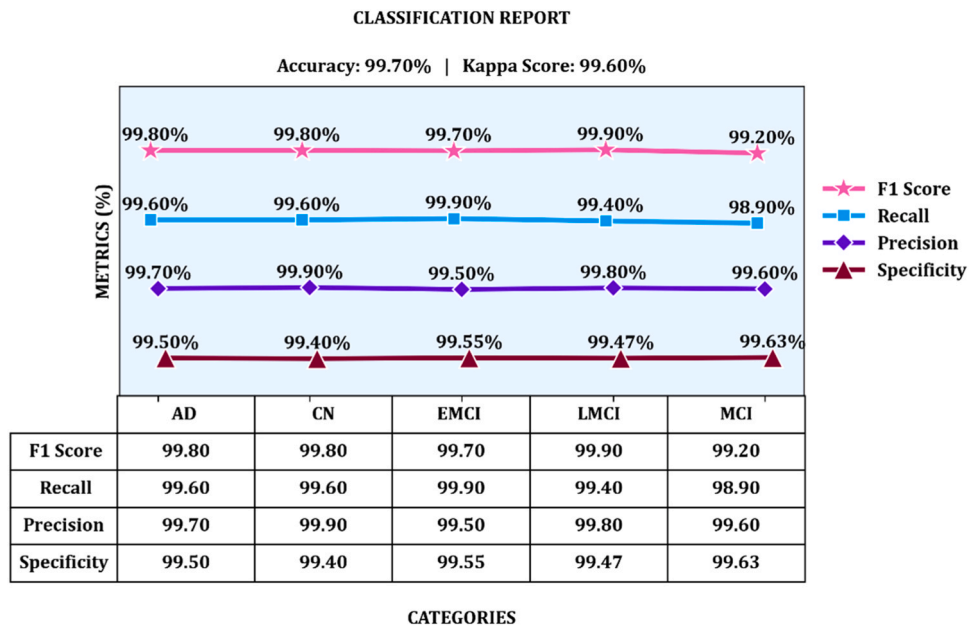


Fig. 12. The classification report of the proposed model on the ADNI dataset.

segmentation model (Improved XceptionNet model) for the tumor region of AD on the OASIS dataset. Three columns make up the illustration, each of which represents a distinct facet of the model's functionality. In the segmentation model for prediction, the classified labelled image is sent, as shown in Column 1. Column 2 displays the ground truth image, which displays the precise locations of the brain tumor as annotated by medical professionals. The brain tumor's core region, as seen by the deep learning model, is displayed in Column 2. Because of its exceptional accuracy, our proposed segmentation model from the test data correctly predicted each region in the original MRI image. Since the trained model had never seen this image, and test data had not been included during training, the proposed segmentation model can predict each region in the image with higher accuracy. With an IoU of 98.57% and a DSC of 98.95% measures for the OASIS dataset and an IoU of 98.87% and a DSC of 98.76% for the ADNI dataset, the

proposed model (Improved XceptionNet) segments the relevant AD region.

4.6. Confusion matrix

It is used to calculate and assess several categorization model metrics. During the testing stage, the model's predictions are divided numerically. Fig. 14 displays the confusion matrix of the proposed model for the OASIS and ADNI datasets, respectively. In the test data, the proposed network's performance in each class is evaluated by a confusion matrix. In the case of four multi-class classifications, Fig. 14 (a) displays the confusion matrix for the OASIS dataset, where one ND participant was incorrectly classified as MD and another as VMD. Overall, this proposed an influential paradigm because, in medical diagnosis, it is better to screen a person for a disease than to rule out a

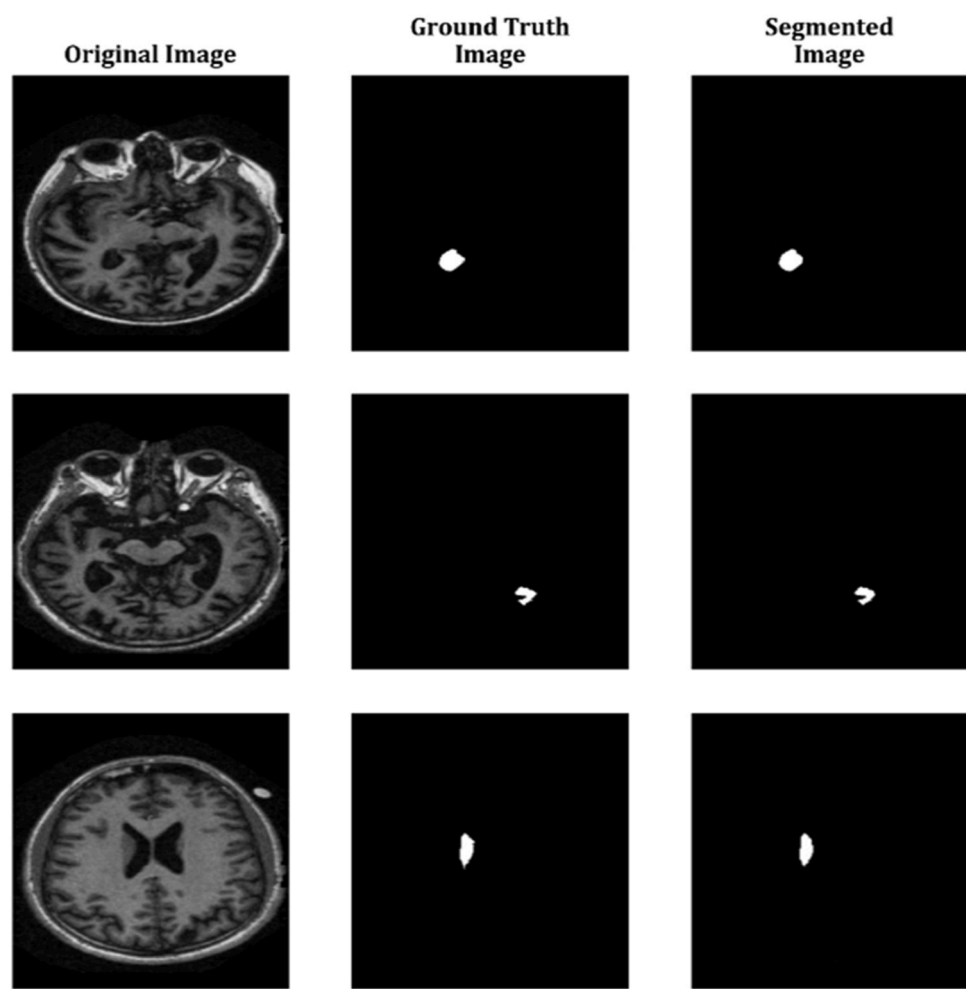


Fig. 13. The segmented results of the proposed model on the OASIS dataset (a) Classified image for prediction, (b) Ground Truth image, and (c) Our segmented image.

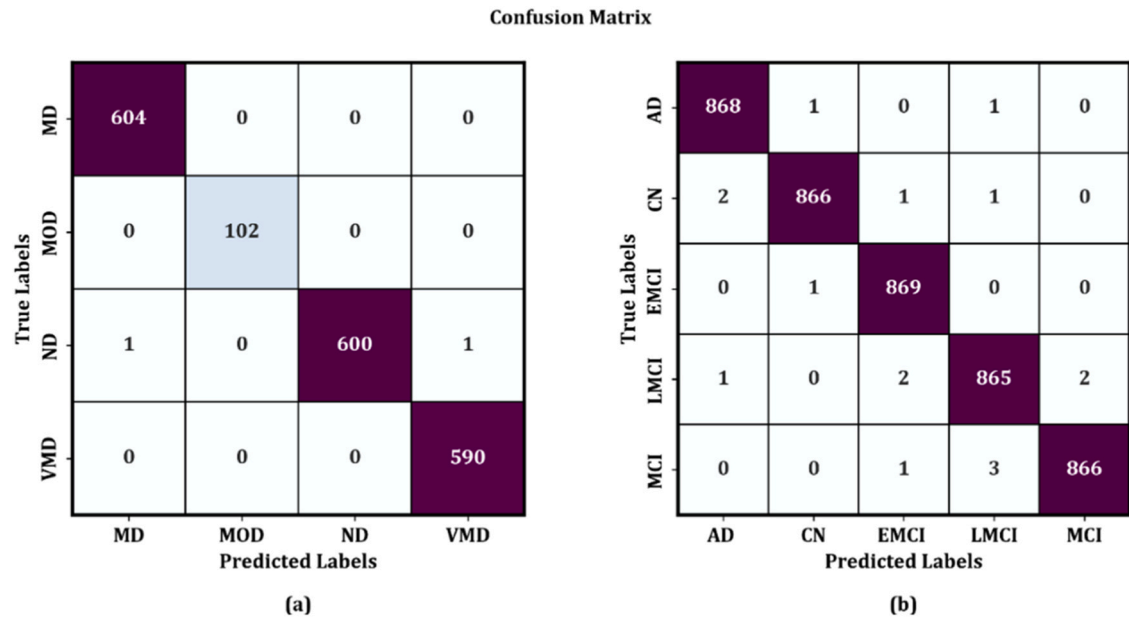


Fig. 14. The proposed model's confusion matrix using test data for (a) the OASIS dataset and (b) the ADNI dataset.

diseased person by making a false negative prediction. [Figs. 15 and 16](#) show the visualized results for the multi-class classification of AD using

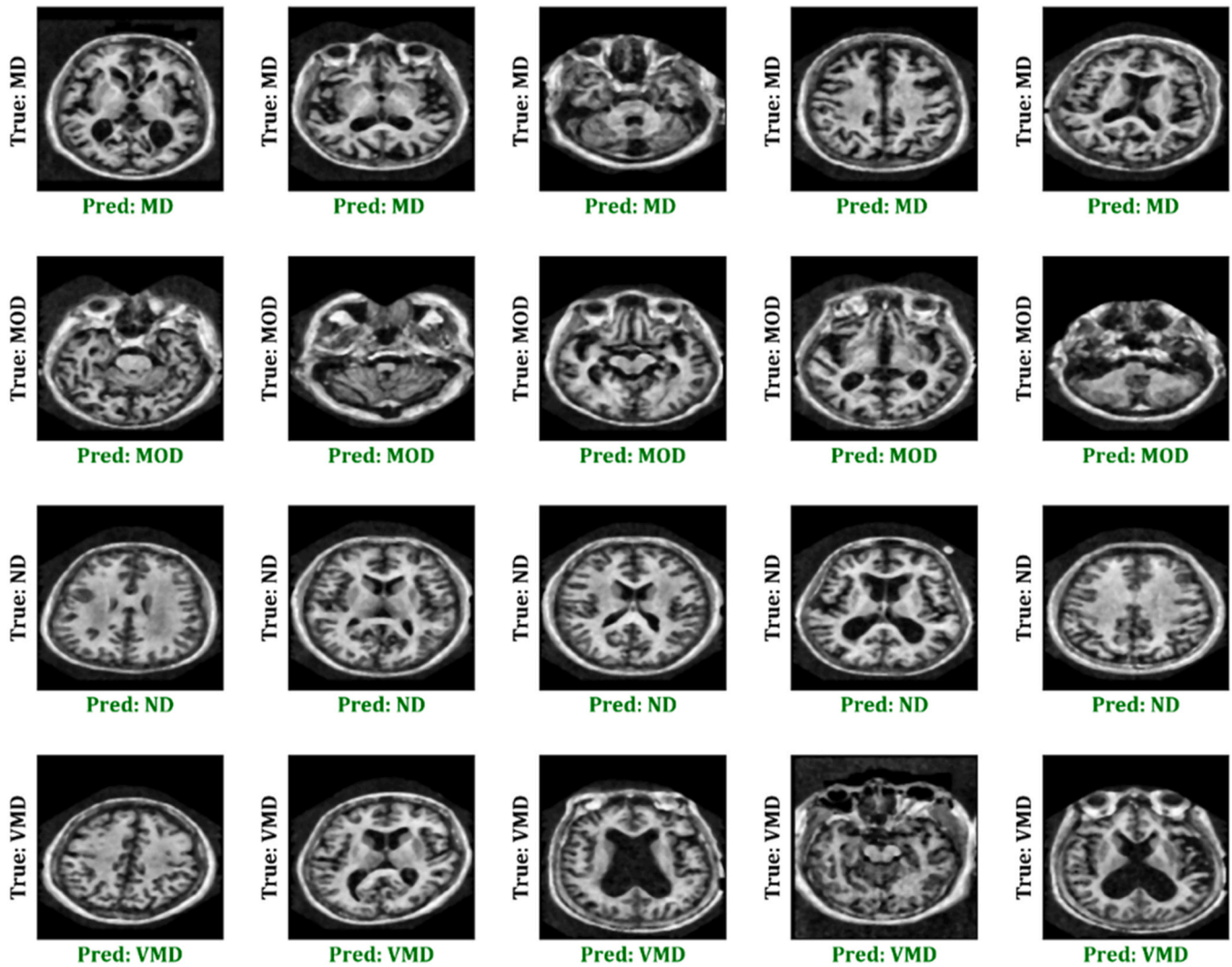


Fig. 15. The visualized classified results of the proposed classification system for the OASIS dataset.

OASIS and ADNI datasets. The proposed model produced average accuracy values of 99.5% and 99.7% for the OASIS and ADNI datasets, respectively. To ascertain whether the expected and actual labels matched, the proposed model was investigated.

4.7. Accuracy and loss graphs

The models' accuracy and loss graphs during training and testing for the OASIS and ADNI datasets are displayed in Figs. 17 and 18. During the training and validation stages, we can determine whether or not over-fitting is present by examining the accuracy/loss graphs of the models. After a specific epoch, the training and validation acc/loss plots in the proposed models for the OASIS dataset (Fig. 17a and Fig. 17b) and ADNI dataset (Fig. 18a and Fig. 18b) continue to overlap. The graphs continue to flatten after a given epoch, indicating that these models have hit their maximum learning capability and are not over-fitting. The majority of learning was typically accomplished by the proposed model for the OASIS and ADNI datasets up to 80 epochs. The proposed models demonstrate similar efficacy in identifying AD patterns and accomplishing successful generalization throughout the fine-tuning process, despite the inherent variations in dataset features, such as size and class distribution.

4.8. ROC analysis

The ROC curve is specifically used to examine the correctness of the proposed classification model for multi-class classification with reliability during an analysis of clinical testing performance. The effectiveness of a classifier is assessed using the area under the curve (AUC) of the ROC curve; generally speaking, a higher AUC indicates that the classifier is more helpful. By using the ROC curve for each class separately and applying our proposed Enhanced Inception ResNetV2 model to the AD data set, we assess its accuracy and applicability. The ROC curve analysis for the OASIS and ADNI datasets is displayed in Fig. 19. For classes MD, MOD, ND, and VMD in the OASIS dataset, the area under the curve (AUC) is 99.42%, 98%, 97.72%, and 97.17%, respectively, as Fig. 19a illustrates. According to Fig. 19b, the AUC for classes AD, CN, EMCI, LMCI, and MCI in the ADNI dataset is 98%, 98%, 97%, 97%, and 98%, respectively. These improvements in AUC demonstrate the high efficacy of the proposed Enhanced Inception-ResNetV2 model utilized in the multi-class AD diagnosis.

4.9. Precision-recall curve

Better discrimination and overall model performance in differentiating between positive and negative examples are indicated by a higher AUPR. All things considered, applying the proposed Enhanced Inception-ResNetV2 model probably results in better classification

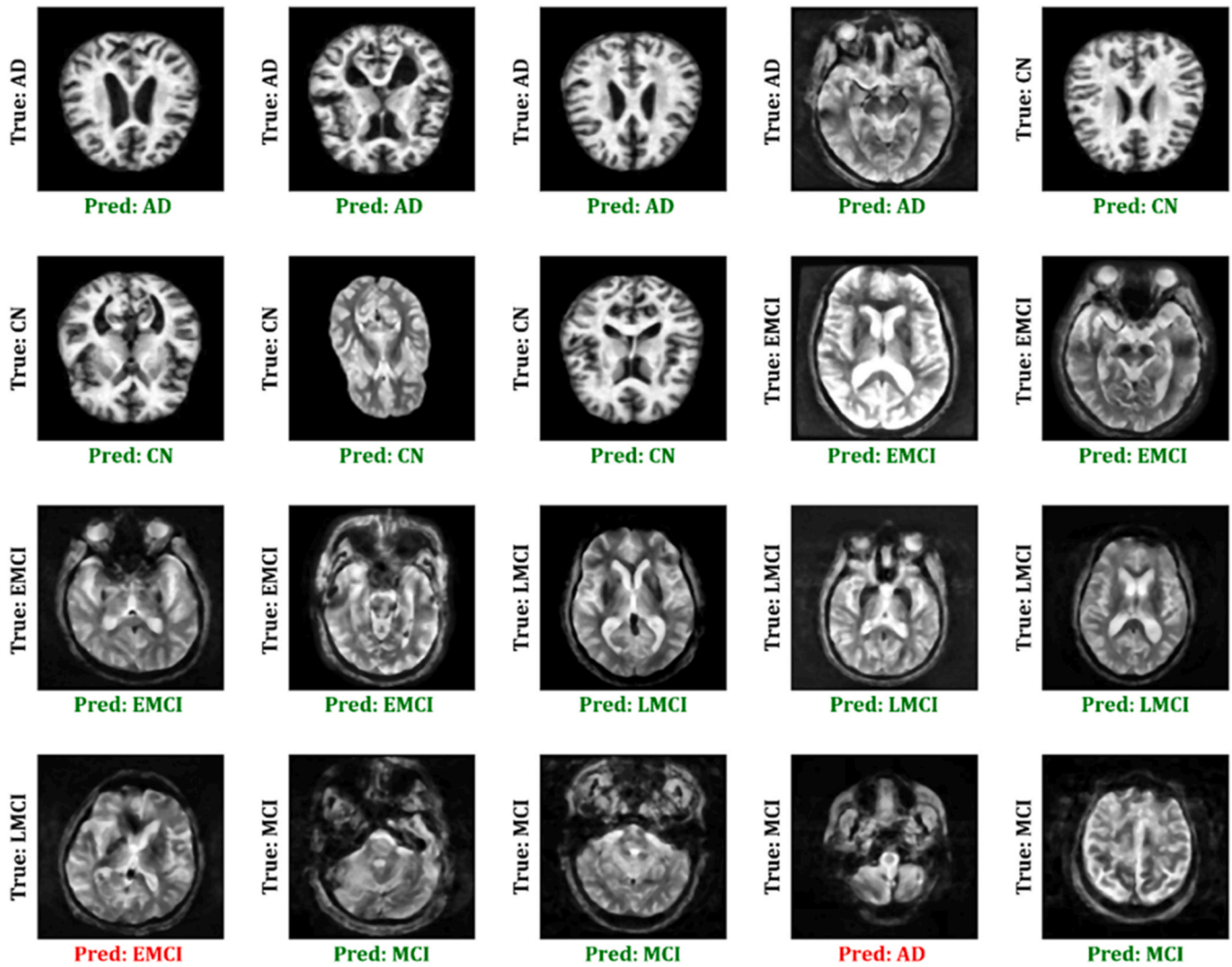


Fig. 16. The visualized classified results of the proposed classification system for the ADNI dataset.

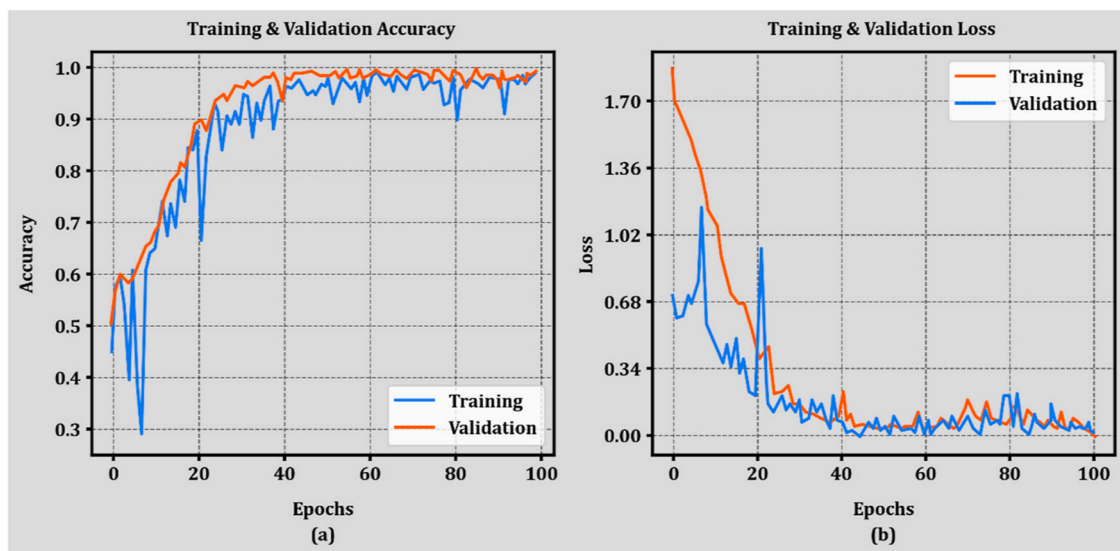


Fig. 17. (a) Accuracy and (b) loss graph on training and testing using the OASIS dataset.

performance and more separability of the anticipated AD classes. Fig. 20 shows the Precision-Recall plot of the proposed classification system for

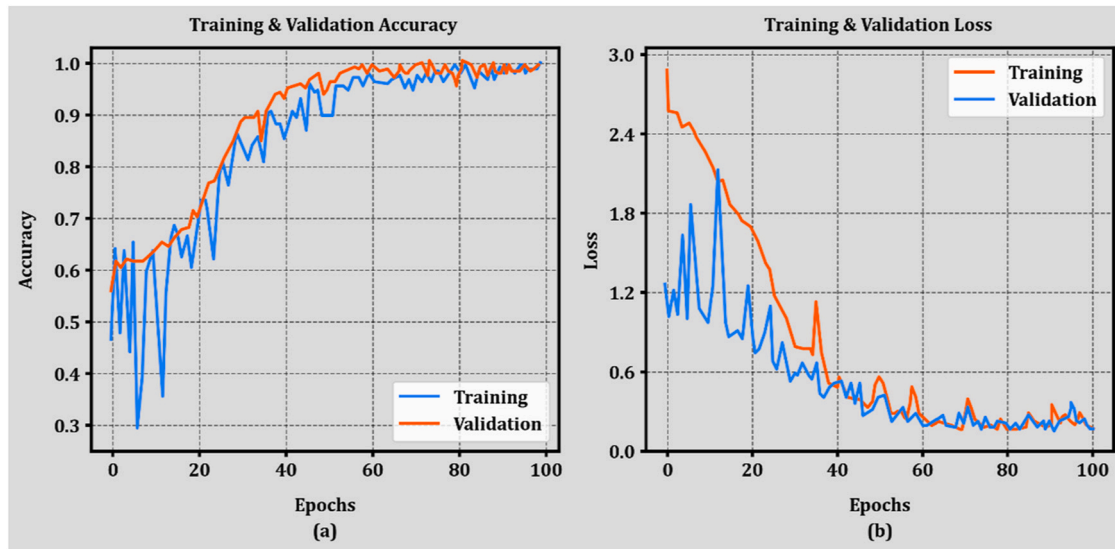


Fig. 18. (a) Accuracy and (b) loss graph on training and testing using the ADNI dataset.

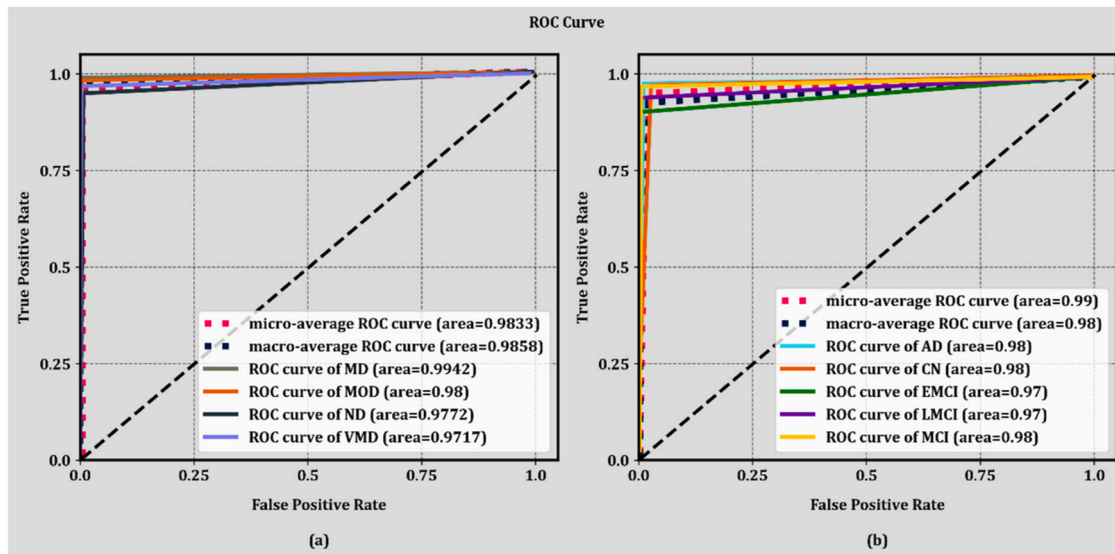


Fig. 19. ROC curve of a proposed classification system for multi-class classification using (a) OASIS dataset and (b) ADNI dataset.

the OASIS and ADNI datasets. In the proposed research, the precision-recall curve illustrates the trade-off between accuracy and recall shows a significant improvement. When there is an imbalance in the data or when the emphasis is on positive examples, the precision-recall curve is very instructive. The precision-recall curve's improvement indicates that the model performs better, properly detecting positive occurrences while reducing the rate of false positives. It also demonstrates that the model achieves more accuracy for a given level of recall. In conclusion, the precision-recall curve significantly improves with the improved model structure using the Adam optimizer, demonstrating the improved discrimination and classification performance of the model for the Alzheimer's diagnostic task using the ADNI and OASIS datasets.

4.10. Performance comparison

The performance comparison of the proposed model with state-of-the-art models on the OASIS and ADNI datasets for multi-class categorization of AD is displayed in Table 5. The proposed methodology outperforms all other approaches with higher accuracy. The framework of

recommended effective segmentation and feature selection techniques makes this feasible. Including irrelevant features may potentially lead to over-fitting and excessive noise, which could produce a subpar model. Additionally, it typically causes the network's performance to deteriorate. This problem is resolved by using the Enhanced Inception ResNetV2 model for classification, which improves the classifier's performance and effectively selects significant features. The proposed (Enhanced Inception-ResNetV2) model outperforms conventional CNN models, with an accuracy of 99.5% for the OASIS dataset and 99.7% for the ADNI dataset. The Inception modules in the proposed classification (Enhanced Inception ResNetV2) model enhance the model's capacity to collect multi-scale features, allowing it to analyze both local and global patterns effectively. Better discrimination between various AD stage categories is made possible by the integration of deep layers from ResNet with inception layers for multi-scale processing. Better discrimination between various AD stage categories is made possible by the integration of deep layers from ResNet with inception layers for multi-scale processing. With a F1-score of 99.52%, recall of 99.57%, and precision of 99.42% for the OASIS dataset and F1-score of 99.68%, recall of 99.48%,

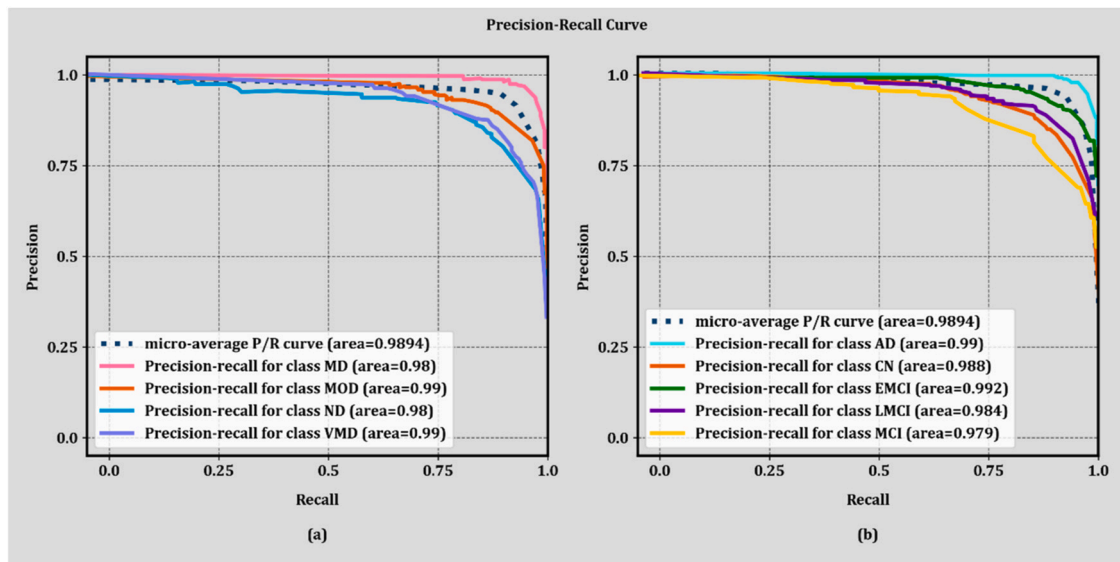


Fig. 20. Precision-Recall curve of proposed classification system for multi-class classification using (a) OASIS dataset and (b) ADNI dataset.

Table 5

Performance comparison of existing methods for OASIS and ADNI datasets for multi-class classification of AD.

Reference	Year	Methods	Datasets	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Nisha et al. (2023)	2023	Hybrid D-OCapNet	OASIS	98.97	98.21	98.31	98.39
			ADNI	99.32	98.93	98.42	98.9
Shaffi et al. (2024)	2024	BEiT+DeiT (Soft)	OASIS	99.06	99.58	97.21	-
			ADNI	98.41	98.73	98.65	-
Matlani (2024)	2024	BiLSTM-ANN	OASIS	98.96	98.43	98.52	98.65
			ADNI	99.22	99.1	99.32	99.42
Heythem et al. (2024)	2024	Customized CNN	ADNI	99.76	99.64	99.62	99.64
Suchitra et al. (2024)	2024	EfficientNetB7	ADNI	98.2	98.95	98.08	98
Kaya et al.	2024	CNN	OASIS	99.53	99.51	99.77	99.63
Proposed Model		Enhanced Inception-ResNetV2	OASIS	99.50	99.42	99.57	99.52
			ADNI	99.70	99.70	99.48	99.68

and precision of 99.70% for the ADNI dataset, our proposed classification model successfully identifies the various AD categories.

The performance of AD segmentation is compared to recent existing models in Table 6. The table shows that the segmentation model that was provided performs noticeably better than earlier deep-learning methods for segmenting the AD region of interest. The proposed approach was able to identify smaller tissue regions and obtain fine details about the boundaries. The proposed segmentation (Improved XceptionNet) model correctly segments the tumor region of AD with an IOU of 98.57% and a DSC of 98.95% for the OASIS dataset and an IOU of 98.87% and a DSC of 98.76% for the ADNI dataset, in contrast to other existing segmentation models for AD segmentation. The proposed XceptionNet splits the feature-filtering and feature-combination process using depth-wise separable convolutions. Because of this, the model is

Table 6

Comparing performance of existing methods for AD segmentation using the ADNI and OASIS datasets.

Reference	Techniques	Datasets	DSC (%)	IOU (%)
Jagadeesh, Kumar (2024)	DFFTU-Net	ADNI	97.02	97.54
Tamilarasi et al., (2024)	Unet	ADNI	98.5	98.7
Rao et al., (2023)	Deep ResUNet	OASIS	98.11	97.5
		ADNI	98.5	97.72
Proposed Model	Improved	OASIS	98.95	98.57
	XceptionNet	ADNI	98.76	98.87

very effective and efficient in capturing spatial features, enabling fine-grained segmentation of AD-affected brain regions. Separable convolutions reduce the computational load, enabling the model to process higher-resolution images more effectively.

4.11. Comparative analysis with state-of-the-art methods

Table 7 (OASIS dataset) and 8 (ADNI dataset) provide a thorough comparison with current state-of-the-art techniques, further validating the efficacy of the proposed hybrid framework. The many deep learning and hybrid methodologies are compiled in these tables, along with the performance metrics that have been reported.

Table 7 shows that a number of recent studies have investigated advanced architectures for the classification of AD, including attention-based DenseNet, Vision Transformers, capsule networks, and hybrid CNN frameworks. For instance, Komal et al., (2025) developed an attention-based DenseNet model that performed well by incorporating self-attention for global feature learning, whereas DuoViT-AD (Trigui and Bezine, 2026) made use of transformer-based dual-scale feature extraction. Similarly, HCapsNet (Narnaware and Mulla, 2025) enhanced feature representation by combining capsule networks with optimization techniques. Although these methods show promising results, many of them involve complex architectures or higher computational requirements.

In contrast, the proposed model achieves an accuracy of 99.5% on the OASIS dataset, outperforming most of the existing methods listed in Table 8. The Enhanced Inception-ResNetV2 model, which effectively captures both local and global features, and PCNN-based feature

Table 7

Performance Comparison for OASIS dataset with state-of-the-art methods.

Reference	Year	Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Gumel et al. (2025)	2025	NASSNet Mobile Network	98	-	-	-	-
Komal et al., (2025)	2025	Attention-based DensNet	97.33	98.51	98.51	98.51	98.51
Trigui and Bezine, 2026	2026	DuoViT-AD	92.15	91.6	92.2	91.9	-
Narnaware and Mulla, 2025	2025	HCapsNet	99.43	95.08	93.52	94.27	98.91
Chaudhari and Khot, 2025	2025	CAdam-RL-DCNN	95.09	93.14	94.52	93.83	95.57
Qian et al., (2025)	2026	MFF-Net	94.19	-	-	86.17	95.92
Alhassan and Altmami, 2025	2026	DEAM-XCN	97.37	97.65	97.09	-	-
Nejad et al. (2026)	2026	Swin Transformer wavelet-GWO	98.12	98.12	98.22	98.16	-
Rani and Baulkani, 2025	2025	ARSA-CA-pre-trained SqueezeNet	98.2	98.8	98.5	98.4	-
Abhaya and Rajkumar, 2025	2025	SpikeGoogle-DeepCNN-SEViT model	95.7	-	97.5	-	95.3
Venkat et al. (2025)	2025	Alzh-Net	99	99	99.25	99	-
Proposed Model		Enhanced inception-ResNet-V2 model-Improved XceptionNet model	99.5	99.42	99.57	99.52	99.37

Table 8

Performance Comparison for the ADNI dataset with state-of-the-art methods.

Reference	Year	Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Soujanya et al. (2025)	2025	FuzzyRes-LeNet-HarFOA	94.475	-	-	-	-
Neethu and Roopa Jayasingh, 2025	2025	MCLSO-SpinalNet	92.6	-	93.6	91.7	93.3
Maganti and Akundi, 2025	2025	Fuzzy Res-LeNet	93.887	-	94.59%	-	94.008
Sathyabhama and Kannan, 2025	2025	EGELU-SZN	95.98	94.16	93.23	-	92.87
Patra et al., (2025)	2025	Transformer-based ResLadderNet	98.92	98.78	95.95	93.18	93.21
Feng et al. (2025)	2025	MCA framework	95.12	-	94.07	-	-
Alsowail and Al-Shehari, 2025	2025	Squeeze Excitation Light Channel Network with a Random Lyrebird algorithm	98.45	-	-	-	-
Jolad et al., (2025)	2025	CATNet	98.2	96.16	98.74	-	-
Chaudhari and Khot, 2025	2026	CAdam-RL-DCNN	96.31	93.87	97.5	95.65	94.06
Qian et al., (2025)	2026	MFF-Net	93.35	-	91.65	91.49	95.11
Alhassan and Altmami, 2025	2026	DEAM-XCN	97.63	98.19	97.07	-	-
Rani and Baulkani, 2025	2025	ARSA-CA-pre-trained SqueezeNet	98.3	98.5	98.3	98.35	-
Proposed Model		Propose model	99.7	99.7	99.48	99.68	99.51

extraction are responsible for the improvement. Moreover, multi-scale depth-wise separable convolutions minimize parameter redundancy while preserving a high degree of discriminative power.

Similarly, the comparative findings on the ADNI dataset (Table 8) show that the proposed model outperforms a number of current methods, including transformer-based ResLadderNet (Patra et al., 2025), DEAM-XCN (Alhassan, Altmami 2025), and CATNet (Jolad et al., 2025), with an accuracy of 99.7%. Although transformer-based models are good at capturing long-range dependencies, they frequently need a lot of processing power and training data. However, the proposed model strikes a fair balance between computational economy and accuracy.

Tables 7 and 8 further highlight the fact that whereas many previous studies concentrate just on classification performance, the proposed research incorporates segmentation and classification into a single framework. By identifying impacted brain regions using the Improved XceptionNet model, this dual-task learning approach not only increases classification accuracy but also enhances interpretability.

Overall, the comparison analysis makes it abundantly evident that the proposed hybrid deep learning methodology maintains computational viability and architectural simplicity while providing better performance than current techniques. This increases the model's suitability for practical clinical applications where efficiency and precision are crucial.

4.12. Computational analysis of the proposed models

Table 9 provides an overview of the proposed hybrid framework's computational performance. The analysis concentrates on training time, inference time per image, and overall inference time throughout the test datasets, all of which are essential components for practical clinical implementation.

The proposed Improved XceptionNet model can produce accurate pixel-level predictions at a reasonable computational cost for the segmentation task, requiring an average inference time of 34.6 ms per image. The model can effectively analyze enormous quantities of MRI

Table 9

Computational analysis of the proposed models.

Model	Task	Training Time	Inference Time per Image (ms)	Total Inference Time (s)
PCNN	Feature Extraction	~1.2 h	10.3	32.5
Enhanced Inception-ResNetV2	Classification	~3.9 h	21.7	74.2
Improved XceptionNet	Segmentation	~4.3 h	34.6	112.8
Proposed Hybrid Model	End-to-End	~5.1 h	66.6	200.5

scans, as evidenced by the overall inference time of 112.8 s for the entire test dataset.

The Enhanced Inception-ResNetV2 model's optimized architecture using depth-wise separable convolutions and residual connections results in an inference time of 21.7 ms per image for classification. Its suitability for fast and precise multi-class AD stage prediction is confirmed by the test dataset's overall classification time of 74.2 s.

Because of its lightweight parallel convolution structure, the PCNN feature extraction module adds very little overhead, with an average processing time of 10.3 ms per image.

These modules are combined sequentially in the overall hybrid framework, which takes about 66.6 ms per image and ~200.5 s to infer the entire test set. This indicates that the proposed approach successfully strikes a balance between high diagnostic accuracy and computational efficiency.

The proposed models have moderate training times; the Enhanced Inception-ResNetV2 takes about 3.9 h, while the Improved XceptionNet takes approximately 4.3 h. These values are acceptable considering the depth of the architectures and the complexity of 3D MRI data.

The computational outcomes unequivocally show that the proposed models are effective enough to be used in computer-aided diagnosis systems. The computational load is greatly reduced without sacrificing performance by the use of depth-wise separable convolutions, feature selection techniques, and optimized architectures. Additionally, the combined segmentation and classification pipeline allows for quick and accurate diagnosis, which makes it appropriate for clinical settings where accuracy and speed are crucial.

4.13. Misclassification analysis for classification and segmentation errors

A thorough misclassification and segmentation error analysis has been performed for both the OASIS and ADNI datasets in order to examine the robustness of the proposed methodology further. A small number of misclassifications and segmentation discrepancies are noted, especially in difficult situations with subtle structural differences, despite the proposed model's excellent overall accuracy. The analysis aids in comprehending the model's shortcomings and possible effects in clinical settings.

4.13.1. Classification error analysis

Tables 10 and 11 show the misclassification results for the OASIS and ADNI datasets, respectively. These tables offer corresponding performance metrics and confidence scores for representative samples when the true label is different from the predicted class.

The majority of misclassifications between closely similar classes, such as Non-Demented (ND), Mild Dementia (MD), and Very Mild Dementia (VMD), may be seen in Table 10 (OASIS dataset). This is mainly due to the minimal visual differences in brain structures across these stages. For instance, there is confusion in feature representation when some ND images are incorrectly classified as MD or VMD with intermediate confidence levels.

Similarly, misclassification between nearby stages, such as EMCI and LMCI, and rarely between AD and EMCI, is shown in Table 11 (ADNI dataset). The main causes of these errors are subtle structural alterations and overlapping imaging characteristics that are hard for even clinical professionals to discern.

Clinical practice may be at risk due to such misclassifications. For example, if an early-stage patient (EMCI) is mistakenly classified as a normal subject, appropriate action may be delayed. If a normal case is predicted to be diseased, needless stress and further diagnostic procedures may result. As a result, even while the model shows a high degree of accuracy, in cases that are close to the edge, careful interpretation is necessary.

4.13.2. Segmentation error analysis

Tables 12 and 13 provide a summary of the segmentation error analysis for the OASIS and ADNI datasets, respectively. In addition to ground truth masks, predicted masks, possible error reasons, and assessment metrics like IoU and DSC, these tables display sample segmentation outcomes.

It is clear from Table 12 (OASIS dataset) that small lesion patches, low contrast boundaries, and minor misalignment between predicted and ground truth masks are the main causes of segmentation errors. Partial detection or under-segmentation can occasionally be seen, especially in areas with low intensity variation.

Similar trends are found in Table 13 (ADNI dataset), where noise, irregular lesion shapes, and intensity variations cause errors.

Table 10
Misclassification analysis of OASIS Dataset.

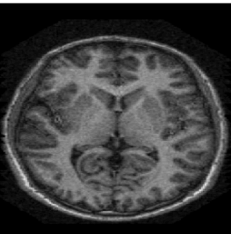
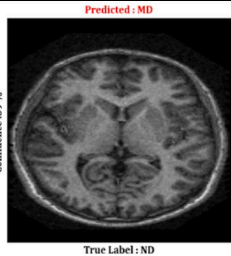
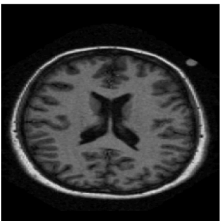
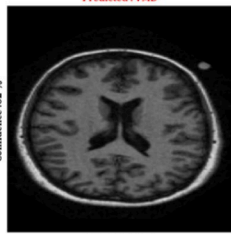
Input Image	True Label	Predicted Image	Predicted Label	Confidence	Performance Metrics
	ND		MD	0.59	Accuracy: 57%, Precision: 51%, Recall: 54%, F1-score: 53%
	ND		VMD	0.62	Accuracy: 60%, Precision: 55%, Recall: 57%, F1-score: 55%

Table 11
Misclassification analysis of ADNI Dataset.

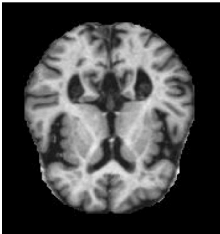
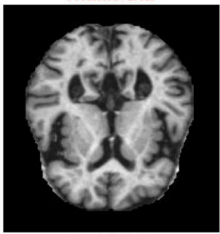
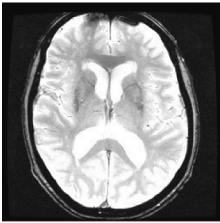
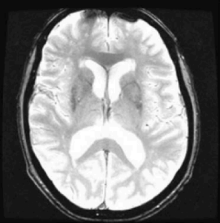
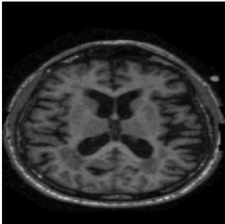
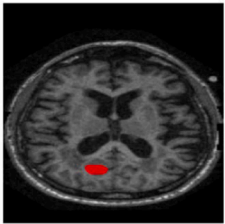
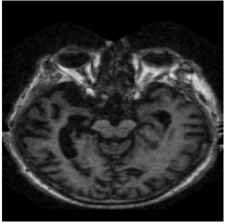
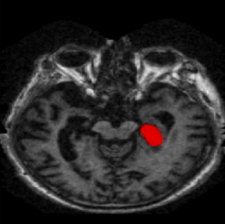
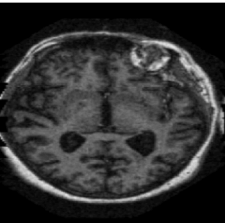
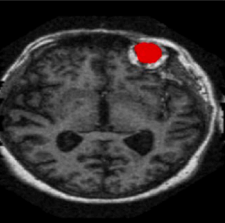
Input Image	True Label	Predicted Image	Predicted Label	Confidence	Performance Metrics
	AD	Predicted : EMCI  True Label : AD	EMCI	0.55	Accuracy: 53%, Precision: 48%, Recall: 52%, F1-score: 50%
	EMCI	Predicted : LMCI  True Label : EMCI	LMCI	0.51	Accuracy: 49%, Precision: 45%, Recall: 49%, F1-score: 47%

Table 12
Segmentation error analysis of OASIS Dataset.

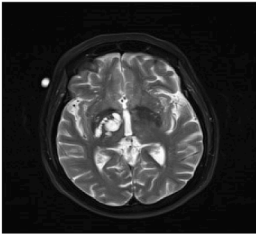
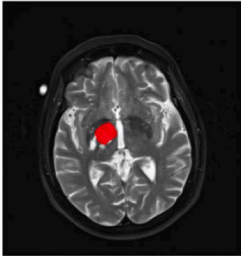
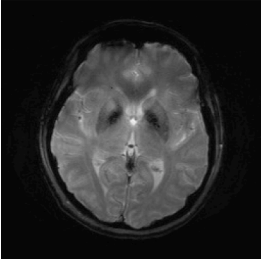
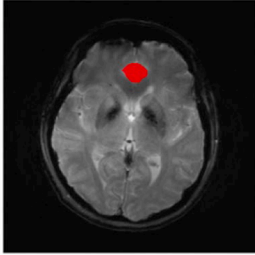
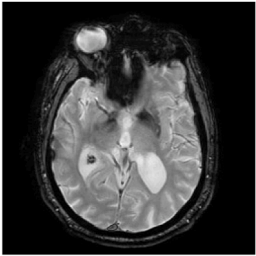
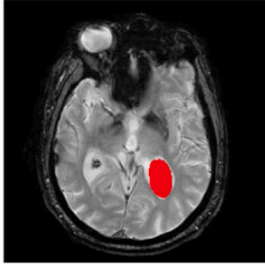
Input Image	Ground Truth Mask	Predicted Mask	Possible Reasons	IoU	DSC
	True MD Region		Small misalignment, partial detection, or low contrast region	65%	67%
	True MOD Region		Slight under-segmentation, image noise, or intensity variation	60%	62%
	True VMD Region		Boundary detection issue, thin region or partial overlap	55%	58%

Particularly in complex brain regions, boundary mismatches and partial overlaps are also noted. In certain instances, these problems somewhat lower the IoU and DSC values.

The precise location of affected brain regions may be impacted by segmentation errors from a clinical standpoint. Under-segmentation

may lead to incomplete identification of diseased areas, while over-segmentation may include healthy tissues, potentially affecting clinical interpretation and treatment planning. The reported errors, however, are small and typically arise in difficult situations with ambiguous boundaries or poor contrast.

Table 13
Segmentation error analysis of ADNI Dataset.

Input Image	Ground Truth Mask	Predicted Mask	Possible Reasons	IoU	DSC
	True AD Region		Small lesion size, intensity variation, or low contrast	65%	67%
	True CN Region		Slight under-segmentation or shape variation	75%	78%
	True MCI Region		Boundary mismatch, partial overlap, or noise	60%	64%

4.14. Discussions

The required measures were taken in this research to guarantee that the segmentation results were reliable and that the experimental settings remained constant. To reduce variances caused by acquisition differences, all input MRI images were pre-processed using the same pipeline, which included resizing, normalization, noise reduction, and contrast enhancement. Additionally, to provide a fair assessment of the proposed models, the same training and testing splits were kept throughout all experiments.

The same optimizer (Adam) was used for both classification and segmentation networks, and hyperparameters like learning rate, batch size, and number of epochs were maintained once optimized to control additional impacting factors. As a result, model performance variability based on training conditions was lessened. To address class imbalance and enhance generalization, data augmentation approaches were utilized consistently across classes.

The segmentation performance was verified using standard evaluation metrics, including IoU and DSC, which are widely used in medical image segmentation studies. The appropriate ground truth annotations found in the datasets were compared to the predicted segmentation masks. To evaluate the alignment between projected regions and actual affected brain areas, segmentation outputs were visually inspected in addition to being quantitatively evaluated.

Additionally, a thorough segmentation error analysis is provided in 4.12, where typical cases and potential causes of errors, such as low contrast regions, tiny lesion size, and boundary ambiguity, are explored. These analyses show areas for future development while also confirming that the proposed segmentation approach yields reliable and consistent

results.

Although they weren't explicitly utilized as model input features, demographic variables like age and sex were taken into account as dataset characteristics in this research. In order to provide a varied collection of brain MRI images, the datasets used in this work—OASIS and ADNI—include people from both genders and a variety of age groups. The majority of the subjects are older since AD is closely connected to aging, and the data used for testing and training reflects this natural pattern.

During model construction, attention was taken to employ balanced samples across all phases of the disease to lessen the impact of bias. However, changes in gender and age-related brain anatomy may still have an impact on the imaging patterns. Particularly at early stages like EMCI and LMCI, these differences can occasionally result in overlapping features between classes.

The model's primary goal in the current work is to learn imaging-based features from MRI data. In order to prevent the model from becoming more complex and dependent on non-imaging data, demographic variables were not specifically included. However, including such information may further improve the performance of the model, particularly in borderline cases. In order to create a more reliable and customized diagnostic system, future research can investigate the integration of clinical and demographic information with imaging features.

4.15. Study limitations and future directions

While the proposed model exhibits encouraging outcomes for segmenting and classifying AD, a number of drawbacks should be noted.

Despite their widespread use in AD research, the OASIS and ADNI

databases have several limitations that should be considered when interpreting the findings. The class imbalance, especially in the MOD category, is one of the primary issues found in the OASIS dataset. This category has substantially fewer samples than other classes, which may have an impact on deep learning models' learning process and result in predictions that are biased toward majority classes.

Data augmentation was used on the MOD class in the training set to lessen this problem. Although this enhances the model's learning and improves the class distribution, supplemented samples may not accurately represent the inherent variability found in actual clinical data. As a result, fluctuations in unseen data may still have a modest impact on the model's performance in this class.

The ADNI dataset is derived from preprocessed and curated images, despite being balanced across all classes. These datasets might not accurately represent real-world clinical settings, where patient diversity, scanner differences, and imaging conditions can add more complexity. This could restrict the model's ability to be directly applied in clinical settings.

The fact that both databases mostly rely on structural MRI images is another drawback. The current study does not take into account important clinical parameters such as multimodal imaging data, cognitive test scores, or demographic information. Adding such data could increase the model's resilience even more.

Despite these drawbacks, the suggested model's overall reliability is enhanced by the use of two distinct datasets and the implementation of augmentation strategies. To further evaluate the model's efficacy, future research will concentrate on assessing it on larger and more varied clinical datasets.

The hybrid model's computing demands are another drawback. The combination of several models (PCNN, Enhanced Inception-ResNetV2, and Improved XceptionNet) raises the overall training time and system complexity, even if the proposed architecture is optimized utilizing depth-wise separable convolutions. Deployment in contexts with limited resources may be difficult as a result. Efficiency can be increased by investigating a lightweight model design or additional optimization.

Furthermore, certain segmentation and classification errors are seen, especially between closely related stages like EMCI and LMCI. Low contrast in MRI images and minute structural variations are the primary causes of these errors. Even while the model works well overall, in some situations, these errors could have an impact on clinical interpretation. This problem might be resolved by implementing efficient feature fusion techniques or attention methods.

Lastly, neither real-time clinical testing nor clinician-in-the-loop validation is included in the current study. The suggested approach is tested in a research environment, and its applicability in clinical workflows has not yet been thoroughly examined. Future research should concentrate on user studies and clinical validation to guarantee its reliability and acceptance in practical applications.

5. Conclusion

AD is a serious neurological disorder that affects a person's cognitive and awareness capacities. Unfortunately, there is presently no all-encompassing treatment for AD. Clinical measures are used to diagnose the disease's progression, but this process is very time-consuming and requires professional skill to identify symptoms. For specialists, early detection is quite difficult until the symptoms are more evident. Therefore, the focus of the research is on developing an effective hybrid deep-learning system for early AD detection. It includes PCNN-based feature extraction, Enhanced Inception ResNetV2-based classification, and Improved XceptionNet-based segmentation for different stages of AD classification. This proposed model achieved success with accuracy rates of 99.5% and 99.7% on two distinct datasets, including OASIS and ADNI, respectively. The test results show that the method suggested in this research outperformed several conventional methods and successfully classified AD on MRI scans. Additionally, the proposed model's

segmentation effectiveness is satisfactory. To help the researchers identify AD more rapidly, we will eventually include more AD variations for classification, such as PMCI, SMCI, and so forth. To increase the precision of our results and provide doctors with a more reliable tool, we will also use other modalities like fMRI, PET, and so on.

Authors' contributions

The author confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation of results, and manuscript preparation.

CRedit authorship contribution statement

V. Srilakshmi: Methodology. **Devarasetty Dr. Prasad:** Data curation. **Chetana Dr. V. Lakshmi:** Conceptualization. **Vinta Dr. SurendraReddy:** Methodology. **Radha Kowtharapu:** Methodology.

Ethics approval

This material is the authors' own original work, which has not been previously published elsewhere. The paper reflects the authors' own research and analysis in a truthful and complete manner.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Abhaya, S.L., Rajkumar, T.D., 2025. Spark-based Alzheimer's disease classification using hybrid spike Google-deep CNN-SEViT model. *Eur. Phys. J.* 140 (8), 756. <https://doi.org/10.1140/epjp/s13360-025-06673-7>.
- Alhassan, A.M., Altmami, N.I., 2025. DEAM-XCN: Alzheimer's disease detection using distributed ensemble activation function based explainable convolutional neural network. *Ain Shams Eng. J.* 16 (10), 103610. <https://doi.org/10.1016/j.asej.2025.103610>.
- Ali, H., Khan, Z., Dai, P., 2026. An attention-guided dual-path residual framework for accurate magnetic resonance imaging-based classification of Alzheimer's disease. *Comput. Electr. Eng.* 129, 110791. <https://doi.org/10.1016/j.compeleceng.2025.110791>.
- Almansour, A.G., Alshomrani, F., Almutairi, A.T., Alalwany, E., Alshuhri, M.S., Alshaari, H., Alfahaid, A., 2025. DA-ViT: deformable attention vision transformer for Alzheimer's Disease Classification from MRI Scans. *Comput. Model. Eng. Sci.* 144 (2), 2395. <https://doi.org/10.32604/cmescs.2025.069661>.
- Alsowail, R.A., Al-Shehri, T., 2025. Multi-stage Alzheimer's disease classification using squeeze excitation light channel network with a random lyrebird: achieving high sensitivity and precision in early detection. *Cogn. Comput.* 17 (4), 128. <https://doi.org/10.1007/s12559-025-10485-3>.
- Basheera, S., Satya Sai Ram, M., 2024. Leung-malik features and adaboost perform classification of Alzheimer's Disease Stages. *IETE J. Res.* 70 (2), 1607–1621. <https://doi.org/10.1080/03772063.2022.2154284>.
- Chaudhari, P.A., Khot, S.S., 2025. Alzheimer's disease prediction using CAdam optimized reinforcement learning-based deep convolutional neural network model. *Biomed. Signal Process. Control* 108, 107968. <https://doi.org/10.1016/j.bspc.2025.107968>.

- Chaudhary, S., Nehra, R., Kumar, K., Dahiya, S., Nandan, D., Kumar, S., 2024. Dementia Alzheimer's disease diagnosis using convolutional neural networks on two-dimensional MRI Slices. *Rev. D'Intell. Artif.* 38 (1). <https://doi.org/10.18280/ria.380109c>.
- Feng, L., Ning, C., Li, Z., Wu, X., Zhao, L., Li, M., 2025. MCA framework: a novel multi-dimensional cooperative analysis framework for Alzheimer's disease diagnosis. *J. King Saud. Univ. Comput. Inf. Sci.* 37 (10), 353. <https://doi.org/10.1007/s44443-025-00344-4>.
- Gautam, P., Singh, M., 2025. A multidimensional U-net architecture for Alzheimer's disease classification using T1 weighted brain structural MRI. *Comput. Electr. Eng.* 128, 110687. <https://doi.org/10.1016/j.compeleceng.2025.110687>.
- Ghassan Al Rahbani, R., Ioannou, A., Wang, T., 2024. Alzheimer's disease multiclass detection through deep learning models and post-processing heuristics. *Comput. Methods Biomech. Biomed. Eng. Imaging Vis.* 12 (1), 2383219. <https://doi.org/10.1080/21681163.2024.2383219>.
- Gumel, A.A., Makama, M.Y., Kazaure, A.A., 2025. Alzheimer disease detection and classification using NASSNet Mobile Network. *Int. J. Sci. Res. Technol.*
- Hasan, M.R., Hossain, A.A., Ullah, S.M.A., 2024. Ensemble ResDenseNet: Alzheimer's disease staging from brain MRI using deep weighted ensemble transfer learning. *Int. J. Comput. Appl.* 46 (7), 539–554. <https://doi.org/10.1080/1206212X.2024.2380648>.
- Hatami, M., Yaghmaee, F., Ebrahimpour, R., 2024. Investigating the potential of reinforcement learning and deep learning in improving Alzheimer's disease classification. *Neurocomputing* 597, 128119. <https://doi.org/10.1016/j.neucom.2024.128119>.
- Heythem, B., Djerioui, M., Beghrich, T., Zerguine, A., Beghdadi, A., 2024. Customized CNN for multi-class classification of brain tumor based on MRI Images. *Arab. J. Sci. Eng.* 1–16. <https://doi.org/10.1007/s13369-024-09284-z>.
- Hu, Z., Wang, Y., Li, Y., 2026. A transformer-based hybrid network for Alzheimer's disease diagnosis via MRI. *Digit. Signal Process.* 168, 105562. <https://doi.org/10.1016/j.dsp.2025.105562>.
- Jagadeesh, B., Kumar, G.A., 2024. Brain tumor segmentation with missing MRI modalities using edge aware discriminative feature fusion based transformer U-net. *Appl. Soft Comput.* 161, 111709. <https://doi.org/10.1016/j.asoc.2024.111709>.
- Jolad, B., Rao, M.V., Kamdi, S.Y., Patil, R.N., Maguluri, L.P., 2025. Alzheimer stages prediction using swinnet for segmentation and transfer learning based CATNet approach with SBO algorithm. *Multiscale Multidiscip. Model. Exp. Des.* 8 (2), 149. <https://doi.org/10.1007/s41939-025-00731-4>.
- Jraba, S., Elleuch, M., Ltfi, H., Kherallah, M., 2024. Alzheimer disease classification using deep CNN methods based on transfer learning and data augmentation. *Int. J. Comput. Inf. Syst. Ind. Manag. Appl.* 16 (3), 17–17.
- Kaya, M., Çetin-Kaya, Y., 2024. A novel deep learning architecture optimization for multiclass classification of Alzheimer's Disease Level. *IEEE Access.* <https://doi.org/10.1109/ACCESS.2024.3382947>.
- Khatri, U., Kwon, G.R., 2024. Diagnosis of Alzheimer's disease via optimized lightweight convolution-attention and structural MRI. *Comput. Biol. Med.* 171, 108116. <https://doi.org/10.1016/j.combiomed.2024.108116>.
- Komal, R., Dhavakumar, P., Rahul, K., Jaswanth, B., Preeth, R., 2025. Hybrid deep learning framework for magnetic resonance imaging-based classification of Alzheimer's disease. *Brain Netw. Disord.* <https://doi.org/10.1016/j.bnd.2025.06.002>.
- Kumar, P.N., Maguluri, L.P., 2024. Deep Learning-based Classification of MRI images for early detection and staging of Alzheimer's Disease. *Int. J. Adv. Comput. Sci. Appl.* 15 (5). <https://doi.org/10.14569/ijcas.2024.0150545>.
- Maganti, S., Akundi, S.H., 2025. A novel hybrid deep learning model for segmentation and uzzy Res-LeNet based classification for Alzheimer's disease. *Neurogenetics* 26 (1), 1–24. <https://doi.org/10.1007/s10048-025-00837-4>.
- Matlani, P., 2024. BiLSTM-ANN: early diagnosis of Alzheimer's disease using hybrid deep learning algorithms. *Multimed. Tools Appl.* 83 (21), 60761–60788. <https://doi.org/10.1007/s11042-023-17867-5>.
- Mohi ud din dar, G., Bhagat, A., Ansarullah, S.I., Othman, M.T.B., Hamid, Y., Alkahtani, H.K., Ullah, I., Hamam, H., 2023. A novel framework for classification of different Alzheimer's disease stages using CNN model. *Electron* 12 (2), 469. <https://doi.org/10.3390/electronics12020469>.
- Narnaware, M.B., Mulla, T.A., 2025. Early detection of Multi-Class alzheimer's disease using hybrid capsule auto encoder model. *Biomed. Signal Process. Control* 110, 108330. <https://doi.org/10.1016/j.bspc.2025.108330>.
- Neethu, M., Roopa Jayasingh, J., 2025. K-Net based segmentation and manta crow light spectrum optimization enabled DNFN for classification of Alzheimer's disease using MRI images. *Multimed. Tools Appl.* 84 (3), 1133–1170. <https://doi.org/10.1007/s11042-024-18967-6>.
- Nejad, A.R., Salimi, F.S.S., Hemmasian, M., Mirzaee, S., Abdiyeva, K., Mousa, R., Hesarak, S., 2026. Multi-class Alzheimer's disease (AD) classification using Swin Transformer wavelet and Gray Wolf Optimization (GWO). *Intell. -Based Med.*, 100362. <https://doi.org/10.1016/j.ibmed.2026.100362>.
- Nisha, A.V., Rajasekaran, M.P., Kottaimalai, R., Vishnuvarthanan, G., Arunprasad, T., Muneeswaran, V., 2023. Hybrid d-capnet: automated multi-class Alzheimer's disease classification in brain mri using hybrid dense optimal capsule network. *Int. J. Pattern Recognit. Artif. Intell.* 37 (15), 2356025. <https://doi.org/10.1142/S0218001423560256>.
- Pallawi, S., Singh, D.K., 2026. LEFF-ViT: a locally enhanced vision transformer framework for accurate alzheimer's disease classification from brain MRI. *Psychiatry Res. Neuroimaging*, 112155. <https://doi.org/10.1016/j.psychresns.2026.112155>.
- Patra, K.J., Mishra, J., Dash, S.K., Mohapatra, S.K., 2025. Enhancing Alzheimer's disease classification using transformer based neuroimaging technique. *J. Transform. Technol. Sustain. Dev.* 9 (1), 10. <https://doi.org/10.1007/s41314-025-00071-8>.
- Qian, Q., Cao, W., Song, M., Feng, Y., Li, Y., Pan, J., 2025. Alzheimer's disease recognition with a multi-scale feature fusion network based on embedding attention maps. *Appl. Soft Comput.*, 114481. <https://doi.org/10.1016/j.asoc.2025.114481>.
- Rahman, A.U., Ali, S., Saqia, B., Halim, Z., Al-Khasawneh, M.A., AlHammadi, D.A., Khan, M.Z., Ullah, I., Alharbi, M., 2025. Alzheimer's disease prediction using 3D-CNNs: Intelligent processing of neuroimaging data. *SLAS Technol.* 32, 100265. <https://doi.org/10.1016/j.slant.2025.100265>.
- Rani, K.E.E., Baulkani, S., 2025. Alzheimer disease classification using optimal clustering based pre-trained SqueezeNet model. *Biomed. Signal Process. Control* 100, 107032. <https://doi.org/10.1016/j.bspc.2024.107032>.
- Rao, B.S., Aparna, M., Harikiran, J., Reddy, T.S., 2023. An effective Alzheimer's disease segmentation and classification using Deep ResUnet and Efficientnet. *J. Biomol. Struct. Dyn.* 1–12. <https://doi.org/10.1080/07391102.2023.2294381>.
- Rao, B.S., Aparna, M., Kolisetty, S.S., Janapana, H., Koteswararao, Y.V., 2024. Multi-class Classification of Alzheimer's disease using deep learning and transfer learning on 3D MRI Images. *Traite du Signal* 41 (3). <https://doi.org/10.18280/ts.410328>.
- Rasheed, Z., Ma, Y.K., Ullah, I., Al Shloul, T., Tufail, A.B., Ghadi, Y.Y., Khan, M.Z., Mohamed, H.G., 2023. Automated classification of brain tumors from magnetic resonance imaging using deep learning. *Brain Sci.* 13 (4), 602. <https://doi.org/10.3390/brainsci13040602>.
- Ravi, V., EA, G., KP, S., 2024. Deep learning-based approach for multi-stage diagnosis of Alzheimer's disease. *Multimed. Tools Appl.* 83 (6), 16799–16822. <https://doi.org/10.1007/s11042-023-16026-0>.
- Sam, F., Qin, Z., Sey, C., Arhin, J.R., Addo, D., Fiasam, L.D., Ayivi, W., Muoka, G.W., 2026. Multisite T1-weighted MRI classification of Alzheimer's disease using 3D-CNN-HSCAM architecture with contrastive domain adaptation. *Biomed. Signal Process. Control* 112, 108686. <https://doi.org/10.1016/j.bspc.2025.108686>.
- Sasikala, T.S., 2024. A CAD system design using iteratively reweighted fuzzy c-means and deep tree training for Alzheimer's disease diagnosis. *Biomed. Signal Process. Control* 88, 105655. <https://doi.org/10.1016/j.bspc.2023.105655>.
- Sasithradevi, A., Baskar, C., Shoba, S., 2024. A HybridOpt approach for early Alzheimer's Disease diagnostics with Ant Lion Optimizer (ALO). *Alex. Eng. J.* 109, 112–125. <https://doi.org/10.1016/j.aej.2024.08.089>.
- Sathyabhama, B., Kannan, M., 2025. An effective deep learning-based automatic prediction and classification of Alzheimer's disease using EGELU-SZN technique. *Neural Comput. Appl.* 37 (9), 6915–6932. <https://doi.org/10.1007/s00521-025-10994-0>.
- Shaffi, N., Viswan, V., Mahmud, M., 2024. Ensemble of vision transformer architectures for efficient Alzheimer's Disease classification. *Brain Inform.* 11 (1), 25. <https://doi.org/10.1186/s40708-024-00238-7>.
- Shankar, V.G., Sisodia, D.S., Chandrakar, P., 2025. Alzheimer's stage progression modeling using graph neural network and MRI biomarkers. *Neural Comput. Appl.* 37 (21), 16825–16847. <https://doi.org/10.1007/s00521-025-11353-9>.
- Shastri, K.A., 2024. Deep Learning-Based Classification of Alzheimer's disease using MRI scans: a customized convolutional neural network approach. *SN Comput. Sci.* 5 (7), 917. <https://doi.org/10.1007/s42979-024-03284-4>.
- Soujanya, R., Syamala, M., Akundi, S.H., 2025. Segmentation and Fuzzy Res-LeNet-based classification for Alzheimer's disease using harmonic fossa optimization. *Biomed. Mater. Devices* 1–31. <https://doi.org/10.1007/s44174-025-00508-w>.
- Suchitra, S., Krishnasamy, L., Poovaraghan, R.J., 2024. A deep learning-based early alzheimer's disease detection using magnetic resonance images. *Multimed. Tools Appl.* 1–22. <https://doi.org/10.1007/s11042-024-19677-9>.
- Tamilaras, M., Kumarganesh, S., Sagayam, K.M., Andrew, J., 2024. Detection and Segmentation of Glioma Tumors Utilizing a UNet convolutional neural network approach with non-subsampled shearlet transform. *J. Comput. Biol.* 31 (8), 757–768. <https://doi.org/10.1089/cmb.2023.0339>.
- Thamizharasi, M., Lakshmi, M., 2025. FEWgat-AGCE: a novel deep learning approach for early Alzheimer's disease detection using hippocampal subfields and cognitive assessments. *Int. J. Mach. Learn. Cybern.* 16 (11), 9165–9192. <https://doi.org/10.1007/s13042-025-02748-0>.
- Trigui, S., Bezine, H., 2026. DuoViT-AD: a dual-scale vision transformer for Alzheimer's disease detection from Brain MRI. *Pattern Recognit. Lett.* <https://doi.org/10.1016/j.patrec.2026.02.033>.
- Venkat, S., Ghodeswar, T., Chavan, P., Narayanasamy, S.K., Srinivasan, K., 2025. MRI-based automated diagnosis of Alzheimer's disease using Alzh-Net deep learning model. *Biomed. Signal Process. Control* 102, 107367. <https://doi.org/10.1016/j.bspc.2024.107367>.
- Waldo-Benitez, G., Padierna, L.C., Ceron, P., Sosa, M.A., 2024. Dementia classification from magnetic resonance images by machine learning. *Neural Comput. Appl.* 36 (6), 2653–2664. <https://doi.org/10.1007/s00521-023-09163-y>.
- Yang, Z., Liu, W., Gan, H., Huang, Z., Zhou, R., Shi, M., 2024. Alzheimer's disease classification based on brain region-to-sample graph convolutional network. *Biomed. Signal Process. Control* 96, 106589. <https://doi.org/10.1016/j.bspc.2024.106589>.
- Zhang, L., Xia, R., Yang, B., Zhang, J., Wang, J., 2024. MSFNet-2SE: a multi-scale fusion convolutional network for Alzheimer's disease classification on magnetic resonance images. *Int. J. Imaging Syst. Technol.* 34 (4), 23112. <https://doi.org/10.1002/ima.23112>.